FLUID PHYSIOLOGY

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Book: Fluid Physiology (Brandis)

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CHAPTER OVERVIEW

1: Introduction

1.1: Properties of Water

Water is one of the two major solvents in the body. It is a remarkable substance with several important properties, in particular, it has:

- A very high molar concentration
- A large dielectric constant
- A very small dissociation constant

Its concentration in biological systems is very high: 55.5 Molar at 37°C (see Box below). This is almost 400 times the concentration of the next most concentrated substance in the body (ie [Na^+] in ECF = 0.14M, [K^+] in ICF = 0.15M). The significance is that water provides an inexhaustible supply of hydrogen ions for the body.

Calculation of Water Concentration

Molecular weight of $H_2O = (1+1+16) = 18$ so one mole is 18 grams

One ml of liquid H_2O weighs about 1 gram (so 1 litre weighs 1,000 grams)

Therefore: $[H_2O] = \frac{1000}{18} = 55.5 moles/liter$

The large dielectric constant means that substances whose molecules contain ionic bonds will tend to dissociate in water yielding solutions containing ions. This occurs because water as a solvent opposes the electrostatic attraction between positive and negative ions that would prevent ionic substances from dissolving. The ions of a salt are held together by ionic forces as defined by Coulomb's Law.

Coulomb's Law

$F = \frac{k \cdot q_1 \cdot q_2}{/} D \cdot r^2$

where:

- F is the force between the two electric charges q_1 and q_2 at a distance r apart
- D is the dielectric constant of the solvent.

The large dielectric constant of water means that the force between the ions in a salt is very much reduced permitting the ions to separate. These separated ions become surrounded by the oppositely charged ends of the water dipoles and become hydrated. This ordering tends to be counteracted by the random thermal motions of the molecules. Water molecules are always associated with each other through as many as four hydrogen bonds and this ordering of the structure of water greatly resists the random thermal motions. Indeed it is this hydrogen bonding which is responsible for its large dielectric constant.

Water itself dissociates into ions but the dissociation constant is very small $K_w = 4.3 \times 10^{-16} mmol/l$ The paradox here is that though this is incredibly small, it has an extremely large effect in biological systems. Why? Because the dissociation produces protons (ie H^+). These are very reactive and have a biologic importance out of all proportion to their minute concentration. (Why? See **Importance of Intracellular pH**)

Physiological Significance of Water's Unusual Properties

Property	Significance
High molar concentration	*Provides inexhaustible supply of H^+*
Large dielectric constant	*Allows Ionic substances to dissolve producing charged species*
Very small dissociation constant	*Produces extremely small but biologically significant $[H^+]$*

Key Point: Water makes it happen!

Water is often treated as though it was just a bland and simple solvent that happens to holds the various solutes in the body. The truth is that it is a solvent with properties unusual enough to allow the situation to occur in the first place.

1.2: Water Movement Across Membranes

1.2.1: Pathways for Water Movement

Oil & water don't mix

Water and lipids are the two major types of solvent in the body. The lipid cell membrane separates the intracellular fluid from the extracellular fluid (as discussed in Section 2.1). Substances which are water soluble typically do not cross lipid membranes easily unless specific transport mechanisms are present. It might be expected that water would likewise not cross cell membranes easily. Indeed, in artificial lipid bilayers, water does not cross easily and this is consistent with our expectation.

. . . but paradoxically, water crosses nearly all the membranes in the body with ease!

Two questions spring immediately to mind:

- How can this be so?
- How does it happen? (ie. What is the route & mechanism by which water crosses membranes?)

The answer to this problem:

Water molecules cross cell membranes by 2 pathways which we can call the lipid pathway & the water channel pathway.

What is the 'lipid pathway'?

This refers to water crossing the lipid bilayer of the cell membrane by diffusion. This initially does not seem to be very credible based on the 'oil & water don't mix' idea BUT it is nonetheless extremely important because this pathway is available in ALL cells in the body.

To express this slightly differently: The 'oil & water don't mix' idea can be quantified as the partition coefficient (ie concentration of water in the lipid phase to the concentration in the aqueous phase). This partition coefficient is as expected, extremely low: about 10-6 which is 1 to a million.

Now there are a couple of other equally important facts to consider:

- the concentration of water in water is extremely high
- the surface area of the cell membrane is very large (relative to the contained volume)

These factors must be included when considering diffusion across the membrane (as quantified by Fick's law of Diffusion) and they significantly counteract the the very low permeability.

The lipid composition of different cell membranes varies so the rate of fluid flow across cell membranes does vary.

What is the 'water channel' pathway?

In some membranes the water flux is very high and cannot be accounted for by water diffusion across lipid barriers. A consideration of this fact lead to the hypothesis that membranes must contain protein which provide an aqueous channel through which water can pass. The water channels have now been found and are discussed below. Flow of water through these channels can occur as a result of diffusion or by filtration.

What other factors are important for the passage of water across membranes?

The above discussion refers to water moving from one side of a lipid barrier to the other and this is relevant to the cell membrane. Other 'membranes' need to be considered; in particular the capillary membrane & the lymphatic endothelial membrane. These are tubular sheets of very many endothelial cells, each with their own cell membrane, but also with a potential pathway for water & solutes existing at the junction of adjacent cells. Similarly all epithelial cell layers can be considered as 'membranes' through which water passes and these also have intercellular pathways.

1.2.2: Capillary Membranes

Water can cross capillary membranes via:

- the intercellular gaps between the endothelial cells

- pores in the endothelial cells special areas where the cytoplasm is so thinned out that it produces deficiencies known as fenestrations.
- diffusion across the lipid cell membranes of the endothelial cells

Intercellular slits in the capillary membrane have a diameter of about 7 nm which is much larger than the 0.12 nm radius of a water molecule. Because the total surface area of the body's capillaries is huge (6,300 m^2) and their walls are thin (1 mm), the total diffusional water flux across the capillaries in the body is very large indeed. (See Section 4.1). Normally this diffusional exchange does not represent any net flow in either direction because the water concentration on both sides of the capillary membrane is the same.

Fenestrations are found only in capillaries in special areas where a very high water permeability is necessary for the function of these areas. A high water permeability is clearly necessary in the glomerular capillaries and water permeability here is very much higher than in muscle capillaries. Other areas with fenestrations are the capillaries in the intestinal villi and in ductless glands.

Water also easily enters the lymphatic capillaries via gaps between the lymphatic endothelial cells. These gaps function also as flap valves and this also promotes forward lymph flow when the capillaries are compressed.

In other areas of the body the water permeability of capillary membranes is quite low. An example is the blood-brain barrier. The capillary endothelial cells here are joined by tight junctions which greatly limit water movement by the intercellular pathway.

1.2.3: Aquaporins: Cell Membrane Water Pores

The presence of specific pores (channels) in the cell membrane has long been predicted but the proteins involved in these water channels have only recently been characterised. At present at least 6 different water channel proteins (named aquaporins) have been found in various cell membranes in humans. These aquaporin proteins form complexes that span the membrane and water moves through these channels passively in response to osmotic gradients. These channel proteins are present in highest concentrations in tissues where rapid transmembrane water movement is important (eg in renal tubules).

Aquaporin 0 is found in the lens in the eye. It has a role in maintaining lens clarity. The gene for this protein is located on chromosome 12.

Aquaporin 1 (previously known as CHIP28) is present in the red cell membrane, the proximal convoluted tubule and the thin descending limb of the Loop of Henle in the kidney, secretory and absorptive tissues in the eye, choroid plexus, smooth muscle, unfenestrated capillary endothelium, eccrine sweat glands, hepatic bile ducts and gallbladder epithelium. The Colton blood group antigen is located on extracellular loop A of aquaporin 1 in red cells. The gene is located on chromosome 7.

Aquaporin 2 is the ADH-responsive water channel in the collecting duct in the inner medulla. Insertion of the channel into the apical membrane occurs following ADH stimulation. The gene is located on chromosome 12.

Aquaporins 3 and 4 are present in the basolateral membrane in the collecting duct. They are not altered by ADH levels. Recently, aquaporin 4 has been found in the ADH-secreting neurones of the supraoptic and paraventricular nuclei in the hypothalamus and it has been suggested that it may be involved in the hypothalamic osmoreceptor which regulates body water balance. (See Section 5.3). The gene for aquaporin 3 is located on chromosome 7.

Aquaporin 5 is found in lacrimal and salivary glands and in the lung. It may be the target antigen in Sjogren's syndrome.

The aquaporins all have a similar topology consisting of 6 transmembrane domains

Aquaporin research is currently an active field. These proteins have been **identified in all living organisms**. New aquaporin inhibitors may prove to be useful diuretic agents. Mercurial compounds used to treat syphilis were noted in 1919 to have a diuretic action. More potent mercurial diuretics were subsequently developed and were once used widely until replaced by less toxic diuretics. These mercurial diuretics act by binding to a specific site on aquaporin 2 with blocking of renal water reabsorption. (See **Section 5.6**)

1.2.4: Effect on Cell Volume

The movement of water across cell membranes is essential for cellular integrity but can cause problems. A small difference in solute concentration results in a very large osmotic pressure gradient across the cell membrane and the cell membranes of animal cells cannot withstand any appreciable pressure gradient. Water movement can eliminate differences in osmolality across the cell membrane but this alone is itself a problem as it leads to alteration in cell volume. Consequently regulation of intracellular solute concentration is essential for control of cell volume.

CHAPTER OVERVIEW

2: Fluid Compartments

2.1: Compartments

The 70 kg 'standard male' contains 42 liters of water - 60% of his body weight. The hypothetical adult female contains 55% of her body weight as water: this lower percent being due to a higher fat content. These figures are standard values which are quoted frequently and are average values.

2.1.1: Variations in Water Content

Variation due to Age

Neonates contain more water then adults: 75-80% water with proportionately more extracellular fluid (ECF) then adults. At birth, the amount of interstitial fluid is proportionally three times larger than in an adult. By the age of 12 months, this has decreased to 60% which is the adult value.

Total body water as a percentage of total body weight decreases progressively with increasing age. By the age of 60 years, total body water (TBW) has decreased to only 50% of total body weight in males mostly due to an increase in adipose tissue.

Variation between Tissues

Most tissues are water-rich and contain 70-80% water. The three major exceptions to this are:

- Plasma: 93% water (& 7% 'plasma solids')
- Fat: 10-15% water
- Bone: 20% water

Variation between Individuals

The variation between individuals in the ratio of TBW to total body weight is quite large but the majority of the variation is due to different amounts of adipose tissue as adipose has a low water content. Differences (between individuals) in the amount of bone and plasma are much smaller. Obese adults have a lower ratio because of the greater amount of adipose tissue. Differences in percent body water between males and females are primarily due to differences in amounts of adipose tissue. For any particular tissue of the body the variation is very much less but any variation that occurs is still mostly due to differences in amount of fat content.

2.1.2: Compartments

The water in the body is contained within the numerous organs and tissues of the body. These innumerable fluids can be lumped together into larger collections which can be discussed in a physiologically meaningful way. These collections are referred to as "compartments"". The major division is into Intracellular Fluid (ICF: about 23 liters) and Extracellular Fluid (ECF: about 19 liters) based on which side of the cell membrane the fluid lies. Typical values for the size of the fluid compartments are listed in the table.

Body Fluid Compartments (70 kg male)

	% of Body Weight	% of Total Body Water	Volume (Litres)
ECF	**27**	**45**	**19**
Plasma	4.5	7.5	3.2
ISF	12.0	20.0	8.4
Dense CT water	*4.5*	7.5	3.2
Bone water	*4.5*	*7.5*	3.2
Transcellular	1.5	2.5	1.0
ICF	**33**	**55**	**23**
TBW	**60%**	**100%**	**42 liters**

2.1.3: Intracellular Fluid

The Intracellular Fluid is composed of at least 10^{14} separate tiny cellular packages. The concept of a single united "compartment" called intracellular fluid is clearly artificial. The ICF compartment is really a "virtual compartment" considered as the sum of this huge number of discontinuous small collections. How can the term intracellular fluid be used as though it was a single body of fluid? The reason is that though not united physically, the collections have extremely important unifying similarities which make the ICF concept of practical usefulness in physiology. In particular, similarities of location, composition and behaviour:

- Location: The distinction between ICF and ECF is clear and is easy to understand: they are separated by the cell membranes
- Composition: Intracellular fluids are high in potassium and magnesium and low in sodium and chloride ions
- Behaviour: Intracellular fluids behave similarly to tonicity changes in the ECF

Because of this physiological usefulness, it is convenient to talk of an idealised ICF as though it were a single real entity. The use of this convention allows predictions to be made about what will happen with various interventions and within limits these are physiologically meaningful.

2.1.4: Extracellular Fluid

A similar argument applies to the Extracellular Fluid. The ECF is divided into several smaller compartments (eg plasma, Interstitial fluid, fluid of bone and dense connective tissue and transcellular fluid). These compartments are distinguished by different locations and different kinetic characteristics. The ECF compositional similarity is in some ways, the opposite of that for the ICF (ie low in potassium & magnesium and high in sodium and chloride).

Interstitial fluid (ISF) consists of all the bits of fluid which lie in the interstices of all body tissues. This is also a virtual fluid (ie it exists in many separate small bits but is spoken about as though it was a pool of fluid of uniform composition in the one location). The ISF bathes all the cells in the body and is the link between the ICF and the intravascular compartment. Oxygen, nutrients, wastes and chemical messengers all pass through the ISF. ISF has the compositional characteristics of ECF (as mentioned above) but in addition it is distinguished by its usually low protein concentration (in comparison to plasma). Lymph is considered as a part of the ISF. The lymphatic system returns protein and excess ISF to the circulation. Lymph is more easily obtained for analysis than other parts of the ISF.

Plasma is the only major fluid compartment that exists as a real fluid collection all in one location. It differs from ISF in its much higher protein content and its high bulk flow (transport function). Blood contains suspended red and white cells so plasma has been called the interstitial fluid of the blood. The fluid compartment called the blood volume is interesting in that it is a composite compartment containing ECF (plasma) and ICF (red cell water).

The fluid of bone & dense connective tissue is significant because it contains about 15% of the total body water. This fluid is mobilised only very slowly and this lessens its importance when considering the effects of acute fluid interventions.

Transcellular fluid is a small compartment that represents all those body fluids which are formed from the transport activities of cells. It is contained within epithelial lined spaces. It includes CSF, GIT fluids, bladder urine, aqueous humour and joint fluid. It is important because of the specialised functions involved. The fluid fluxes involved with GIT fluids can be quite significant. The electrolyte composition of the various transcellular fluids are quite dissimilar and typical values or ranges for some of these fluids are listed in the Table.

The total body water is divided into compartments and useful physiological insight and some measure of clinical predictability can be gained from this approach even though most of these fluid compartments do not exist as discrete real fluid collections.

2.1.5: Functional ECF

The water in bone and dense connective tissue and the transcellular fluids is significant in amount but is mobilised much more slowly then the other components of the ECF. The remaining parts of the ECF are called the functional ECF. The ratio of ICF to ECF is 55:45.

The functional ECF is more important when considering the effects of acute fluid interventions and the ratio of ICF to functional ECF is 55:27.5 (which is 2:1). (See Section 8.1 for discussion of acute fluid infusions).

Typical Electrolyte Concentrations in Some Transcellular Fluids (in mmol/l)

	[Na^+]	**[K^+]**	**[Cl^-]**	**[HCO_3^-]**
Saliva	20-80	10-20	20-40	20-60
Gastric juice	20-100	5-10	120-160	0
Pancreatic juice	120	5-10	10-60	80-120
Bile	150	5-10	40-80	20-40
Ileal fluid	140	5	105	40
Colonic fluid	140	5	85	60
Sweat	65	8	39	16
CSF	147	3	113	25

2.2: Measurement of Compartment Volumes

2.2.1: The Dilution Principle

Compartment volumes are measured by determining the volume of distribution of a tracer substance. A known amount of a tracer is added to a compartment. The tracer concentration in that compartment is measured after allowing sufficient time for uniform distribution throughout the compartment. The compartment volume is calculated as:

Volume = Amount of tracer / Concentration of tracer

Ideally, the tracer should have certain properties (see box)

Properties of an Ideal Tracer

The tracer should:

- be nontoxic
- be rapidly and evenly distribute throughout the nominated compartment not enter any other compartment.
- not be metabolised
- not be excreted (or excretion is able to be corrected for) during the equilibration period
- be easy to measure
- not interfere with body fluid distribution

If the tracer is excreted in the urine, then the loss can be determined and corrections made in the calculation. If the tracer is metabolised, a series of measurements can be made and assuming exponential decline (first order kinetics), the volume of distribution can be determined by extrapolation back to zero time.

2.2.2: Total Body Water

This is estimated by measuring the volume of distribution of isotopes of water. Tritium oxide (THO) is used because it is a weak beta emitter making it easy to measure in a liquid scintillation counter. Rapid mixing of tritiated water throughout all compartments occurs during a 3 to 4 hour equilibration period. Results are accurate and reproducible to within 2 percent.

2.2.3: Extracellular Fluid

Tracers used fall into 2 groups:

- Ionics (eg 82Br, $^{35}SO_4$, chloride isotopes)
- Crystalloids (eg Inulin, mannitol)

The ionic tracers are small and distribute throughout the ECF but there is some entry into cells. ECF will be over-estimated with these tracers

The crystalloids are larger and less diffusable throughout the ECF. They do not enter cells but the lack of full ECF distribution results in a low estimate of ECF.

What is measured is not the true ECF so it is conventional to refer to the compartment measured not as ECF but as a space defined by the tracer used and the equilibration time (eg 20 hour bromide space).

Measurements indicate that the ECF can be modelled as consisting of:

- a rapidly equilibrating pool ("functional ECF") which makes up about 27 to 30% of total body water (This rapid pool represents plasma and most of the ISF)
- a slowly equilibrating pool (24 hours) which makes up 15% of total body water. (This slow pool mostly represents the water of dense connective tissue and bone and some of the transcellular fluid)

2.2.4: Plasma Volume

Measurement of plasma volume requires a tracer which is mostly limited to this compartment and this is achieved by using a tracer which binds to albumin.

The tracers used are the azo dye known as **Evan's blue** (or T1824) which binds avidly to albumin, or **radio-iodine labelled serum albumin (RISA)**. Distribution is rapid but no equilibrium is reached because of continuous disappearance of albumin from the vascular space. This problem is overcome by using serial measurements and plotting the disappearance curve of the label. This is a first order process (ie exponential decline) which gives a straight line when plotted on a logarithmic scale. Extrapolation back to zero time allows estimation of the virtual concentration at this time. The volume is determined via the dilution principle using this concentration at zero time. As the concentration of the tracer is determined in a plasma sample, the measured volume of distribution is the plasma volume.

2.2.5: Blood Volume

The tracer is the patient's own red cells which are tagged with radio-chromium (Cr^{51}-red cells). Typically a 10 ml sample of the subject's blood is incubated with a sodium chromate solution, the label is taken up by the red cells and any excess in the solution is removed by dilution and centrifuging with removal of fluid. The labelled red cells are centrifuged, resuspended in saline and infused into the patient. The volume of distribution (VD) is determined after about 30 minutes. As the radioactive label distributes throughout the whole intravascular compartment, the measured VD is the blood volume (rather than the red cell volume). The distribution of the label is not uniform because the haematocrit is different in different parts of the circulation. It is usual therefore to measure the amount of the label in a red cell sample and therefore to directly measure the red cell volume.

Plasma volume or red cell volume can be determined indirectly if the blood volume and haematocrit (Hct) are known.

Formulae for Blood Volume

- $\text{Blood Volume} = \text{Plasma volume} \times \frac{100}{100-Hct}$
- $\text{Blood Volume} = \text{Red cell volume} \times \frac{100}{Hct}$

(where Hct = Haematocrit)

As mentioned previously, there are several problems in estimating an average or 'whole body' haematocrit:

- Haematocrit measured in the laboratory overestimates true haemotocrit because about 4 to 8% of the plasma remains trapped with the red cells in the tube
- Blood from capillaries has a lower haematocrit then in larger vessels because of axial streaming of red cells. (Haematocrit in muscle capillaries is typically only 0.20 !)
- Large vein haematocrit is higher then in arteries because the various reactions in the red cell due to carbon dioxide transport lead to an increase in the number of particles intracellularly and an osmotic increase in water content

Accounting for these effects, the whole body haematocrit can be estimated as about 91% of large vein haematocrit and this value should be substituted in the equations.

2.2.6: Other Major Compartments

Interstitial Fluid

There is no tracer which are distributed only throughout this compartment. ISF is determined indirectly as the difference between concurrently measured ECF & plasma volumes. Measurement error is the sum of the errors of the two individual measurements and can be significant.

Intracellular Fluid

There is no tracer available so ICF is measured indirectly as the difference between concurrently measured total body water and ECF. The volume of ICF decrease with increasing age and this accounts for most of the age-related decline in total body water.

Transcellular Fluids

There is no tracer for the measurement as a whole of the myriad components of transcellular water. Methods exist for the estimation of the various components individually.

2.3: Osmolarity and Tonicity

Some definitions are necessary first to help us in our discussion of fluid and electrolyte conditions.

Useful Definitions

- **Mole** - A mole is the amount of a substance that contains the number of molecules equal to Avogadro's number. The mass in grams of one mole of a substance is the same as the number of atomic mass units in one molecule of that substance (ie the molecular weight of the substance expressed as grams). The mole (symbol: mol) is the base unit in the SI system for the amount of a substance
- **Avogadro's number** - this is the number of molecules in one mole of a substance (ie 6.022×10^{23})
- **Molality** of a solution is the number of moles of a solute per kilogram of solvent
- **Molarity** of a solution is the number of moles of solute per litre of solution
- **Osmole** - This is the amount of a substance that yields, in ideal solution, that number of particles (Avogadro's number) that would depress the freezing point of the solvent by 1.86K
- **Osmolality** of a solution is the number of osmoles of solute per kilogram of solvent
- **Osmolarity** of a solution is the number of osmoles of solute per litre of solution

Osmolality is a measure of the number of particles present in solution and is independent of the size or weight of the particles. It can be measured only by use of a property of the solution that is dependent solely on the particle concentration. These properties are collectively referred to as the *colligative properties*.

Colligative Properties

- vapour pressure depression
- freezing point depression
- boiling point elevation
- osmotic pressure

The osmotic pressure is the hydrostatic (or hydraulic) pressure required to oppose the movement of water through a semipermeable membrane in response to an 'osmotic gradient' (ie differing particle concentrations on the two sides of the membrane).

Serum osmolality can be measured by use of an osmometer or it can be calculated as the sum of the concentrations of the solutes present in the solution. The value measured in the laboratory is usually referred to as the *osmolality*. The value calculated from the solute concentrations is reported by the laboratory as the *osmolarity*. The Osmolar gap is the difference between these two values. The two values usually don't match exactly for various: there are a number of formulas that can be used and they all give slightly different results; the formulas typically use the concentrations of only 3 solutes (Na, glucose, urea) in the calculation so contributions from abnormal small MW uncharged substances will be missed so the calculated value will be low; use of osmometers that use the vapour pressure method are unreliable in the presence of volatile chemicals.

Tonicity is a term used frequently in a medical context. It is also a term which is frequently misunderstood as it is defined in at least three different ways. The most rigorous & useful definition is:

Tonicity is the *effective osmolality* and is equal to the sum of the concentrations of the solutes which have the capacity to exert an osmotic force across the membrane.

The key parts are *effective* and *capacity to exert*. The implication is that tonicity is less then osmolality. How much less? Its value is less then osmolality by the total concentration of the ineffective solutes it contains. Why are some solutes effective and others ineffective?

Consider this experiment: Imagine a glass U-tube which contains two sodium chloride solutions which are separated from each other by a semipermeable membrane in the middle lowest part of the U-tube (see figure below). The membrane is permeable to water only and not to the solutes (Na^+ and Cl^-) present. If the total particle concentration (osmolality) of Na^+ and Cl^- on one side of the membrane was higher than the other side, water would move through the membrane from the side of lower solute concentration (or alternatively: higher H_2O concentration) to the side of higher solute concentration. Water (the solvent) moves down its concentration gradient.

U-Tube Experimental Setup

coming . . .

If the water levels were different in the two limbs of the U-tube at the start of the experiment then:

- What would be the equilibrium situation as regards particle (ie solute) concentration on the two sides of the membrane?
- What would the difference (if any) be in the heights of the water levels on the two sides of the membrane?
- Is the equilibrium condition reached when the particle concentration (ie osmolality) is equal on the two sides of the membrane?

The answer to the last question is no because this neglects the probable difference in the height of the water columns in the two limbs. This height difference is a hydrostatic (or hydraulic) pressure difference and this provides an additional force which must be accounted for in the balancing of forces needed to reach an equilibrium. (An equilibrium is present when there is no net water movement across the membrane.)

The equilibrium would occur when this net hydrostatic pressure is balanced by the remaining difference in osmolality between the two solutions. This osmolality difference results in an osmotic force which tends to move the water in the opposite direction to the hydrostatic pressure gradient. Equilibrium is when these opposing forces are equal.

Now consider what would happen in the above situation if the membrane was changed to one which was freely permeable both to the water and to the ions (sodium & chloride) present. Now none of the particles present has the capacity to exert an osmotic force across the membrane. At equilibrium there is no difference in the fluid levels in the two limbs of the U-tube because the particles present will move across the membrane until the concentration gradients for Na^+ or Cl^- are eliminated. The osmolality is now the same on both sides of the membrane. At equilibrium there will be no hydrostatic gradient either.

The conclusion is that if the membrane allows certain solutes to freely cross it, then these solutes are totally ineffective at exerting an osmotic force across this membrane and this must be corrected for when considering the particle concentrations across the membrane. Tonicity is equal to the osmolality less the concentration of these ineffective solutes and provides the correct value to use.

Osmolality and Tonicity: Relationship to Membrane

Osmolality is a property of a particular solution and is independent of any membrane.
Tonicity is a property of a solution in reference to a particular membrane.

It is strictly wrong to say this or that fluid is isotonic with plasma - what should be said is that the particular fluid is isotonic with plasma in reference to the cell membrane (ie the membrane should be specified.) By convention, this specification is not needed in practice as it is understood that the cell membrane is the reference membrane involved.

From a cell's viewpoint, it is net osmolar gradient across the cell membrane at any moment that is important. Tonicity (and not osmolality) is important for predicting the overall final outcome (the equilibrium state) of a change in osmolality because it allows for those solutes which will cross the membrane. All the cells in the body (with a few exceptions eg cells in the hypertonic renal medulla) are in osmotic equilibrium with each other. Movement of water across cell membranes occurs easily and rapidly and continues until intracellular and extracellular tonicities are identical. If water can cross the membrane faster than the ineffective solute can cross then the effect of an abrupt change in extracellular osmolality may be initially and temporarily different from that predicted from the acute tonicity change alone.

If a hyperosmolar solution was administered to a patient, this would tend to cause water to move out of the cell. However if the solute responsible for the hyperosmolality was also able to cross cell membranes it would enter the cell, increase intracellular osmolality and prevent this loss of intracellular fluid. This is the situation with hyperosmolality due to high urea concentrations as urea crosses cell membranes relatively easily.

Hyperglycaemia in untreated diabetics results in ECF which is both hyperosmolar and hypertonic (as compared to the normal situation) as glucose cannot easily enter cells in these circumstances. Water moves out of the cells until the osmolar gradient is abolished.

In some situations, a more operational definition of tonicity is used to explain the term: though not incorrect this explanation is less versatile and rigorous than the one discussed above. This is based on the experiment of immersing red cells in various test solutions and observing the result. If the red cells swell and rupture, the test solution is said to be hypotonic compared to normal plasma. If the red cells shrink and become crenated, the solution is said to be hypertonic.

If the red cells stay the same size, the test solution is said to be isotonic with plasma. The red cell membrane is the *reference membrane*. Red cells placed in normal saline (ie 0.9% saline) will not swell so normal saline is said to be isotonic. Haemolysis does not occur until the saline solution is less then 0.5%. The point about this definition of tonicity is that it is qualitative and not quantitative. It does imply that permeant solutes will be ineffective because it is essentially a test against a real membrane.

A major physiology text (Ganong 16th ed., 1993) defines tonicity as a term used to *describe the osmolality of a solution relative to plasma* (as in hypotonic, isotonic or hypertonic). This less rigorous definition is wrong as it does not cover the full sense in which the term tonicity is used. Ganong argues that an infusion of 5% dextrose is initially isotonic but that when the glucose is taken up and metabolised by cells, the overall effect is of infusing a hypotonic solution. This is really a problem with his definition. More correctly, one would say that the 5% dextrose is initially isosmolar with plasma (and this avoids haemolysis). Glucose is a permeant solute in the non-diabetic and can easily enter cells. When infused, the 5% dextrose is very hypotonic (with reference to the cell membrane) despite being isosmolar. Water does not leave the cells initially (and haemolysis does not occur) because there is no osmolar gradient across the cell membrane. The solution is however hypotonic and when the glucose enters cells water does also. If insulin is not present, this movement of glucose does not occur. In this latter case, the solution is isosmolar before infusion and can be considered isotonic after infusion as well.

The particular problem with this definition is that it does not distinguish *tonicity* from *osmolality* as it makes no recognition of whether the available solutes are permeant (and thus 'ineffective') or non-permeant (and thus 'effective') with respect to a particular membrane as in the example of the 5% glucose which is isosmolar but hypotonic. The definition really doesn't add much more then could be achieved by the terms hypo- and hyper-osmolality. Of course, the referencing to actual plasma osmolality means this definition is effectively the same as the 'red cell test' definition, while obscuring the fact that tonicity is referenced to the cell membrane.

Note that tonicity is defined in several ways which don't all have exactly the same meaning. This is confusing. The definition based on tonicity as the effective osmolality is best.

Comparison of Different Definitions of Tonicity

- Effective osmolality - The best definition as it accounts for permeant solutes and is quantitative.
- The red cell test - A practical qualitative definition that emphasises the requirement that tonicity is defined in reference to a membrane.
- Comparison with osmolality of plasma - Does not account for permeant solutes, and not quantitative.

A final point here regarding the meaning of the term "osmotic pressure".

Consider again the U-tube experiment but pure water on one side and a test solution of unknown osmolality on the other side of a semipermeable membrane which is permeable only to water. Water will move into the test solution. What would happen if further amounts of the test solution were added before any movement of water had occurred? An equilibrium situation would be reached at which the hydrostatic pressure (ie difference in fluid heights in the two limbs of the U-tube) on the test solution side of the membrane would balance the osmotic tendency for water to move across the membrane into the test solution.

At this equilibrium point, the hydrostatic pressure is a measure of the osmotic tendency in the test solution: indeed the opposing hydrostatic pressure needed to balance the osmotic forces is usually referred to as the osmotic pressure.

There would be practical difficulties in performing this experiment with body fluids as the test solution as the osmotic pressure to be measured is over 7 atmospheres and an extremely long-limbed U-tube would be necessary! Alternatively, the pressure could be supplied from a piston or a compressed gas source rather than a column of fluid.

2.4: Colloid Osmotic Pressure

2.4.1: Colloids

Definitions

Colloids is a term used to collectively refer to the large molecular weight (nominally MW > 30,000) particles present in a solution. In normal plasma, the plasma proteins are the major colloids present.

As the colloids are solutes they contribute to the total osmotic pressure of the solution. This component due to the colloids is typically quite a small percent of the total osmotic pressure. It is referred to as ***colloid osmotic pressure*** (or sometimes as the ***oncotic pressure***).

In plasma, the oncotic pressure is only about 0.5% of the total osmotic pressure. This may be a small percent but because colloids cannot cross the capillary membrane easily, oncotic pressure is extremely important in transcapillary fluid dynamics.

Measurement

Oncotic pressure can be easily measured in the laboratory with instruments called oncometers.

The principle is to have 2 chambers which are enclosed and separated from each other by a semi-permeable membrane which is:

- permeable to water and small MW substances, *but*
- not permeable to molecules with a MW greater then 30,000 (ie colloids)

Relative to this membrane, the colloids are the only *effective* solutes present. The reference chamber contains isotonic saline and the test solution is added to the sample chamber. If the test solution contains colloids, water moves from the reference chamber to the sample chamber. The decrease in pressure in the test chamber is detected by a pressure transducer (strain gauge) mounted between the two chambers. Modern oncometers can provide accurate results with samples as small as 50 microlitres in a couple of minutes.

2.4.2: van't Hoff Equation

Osmotic pressure and oncotic pressure can be *measured* by suitable instruments. They can also be determined for a solution by appropriate substitutions in the van't Hoff equation. This equation expresses the relationship between solute concentration and osmotic pressure for *ideal* solutions. (An ideal solution is a solution with thermodynamic properties analogous to those of a mixture of ideal gases - see a chemistry text for more details.)

The van't Hoff Equation

$$\text{Osmotic pressure} = \boldsymbol{n} \times \frac{\boldsymbol{c}}{\boldsymbol{M}} \times \boldsymbol{RT}$$

where:

- n is the number of particles into which the substance dissociates (n = 1 for plasma proteins)
- c is the concentration in G/l
- M is the MW of the molecules
- c/M is thus the molar concentration of the substance (mol/l)
- R is the universal gas constant
- T is the absolute temperature (K)

As an example, if values are substituted in this equation for a typical plasma sample:

- $T = 310K$ (i.e. temp of 37°C)
- $R = 0.082 \frac{litre \cdot atm}{K \cdot mol}$
- $n = 1$ (for plasma proteins as they do not dissociate)

and:

- Multiplying by 0.001 to convert from Osmoles to mOsmoles
- Multiplying by 760 to convert the result from atmospheres to mmHg
- Multiplying by 280 to convert the osmotic pressure per mOsm/kg to a value for plasma with an osmolality of 280 mOsm/kg

then:

$$\text{Total plasma osmotic pressure} = 1 \times 0.082 \times 310 \times 0.001 \times 760 \times 280 = 5409 mmHg$$

For a plasma osmolality of 280 mOsm/kg at 37° C, total osmotic pressure is about 5409mmHg (ie about 7.1 atmospheres!)

Each mOsm/kg of solute contributes about 19.32mmHg to the osmotic pressure

Now consider the case of plasma proteins alone and calculate the colloid osmotic (oncotic) pressure.

Using typical values for concentration & MW of the plasma proteins, the protein concentration is about 0.9 mOsmol/kg which predicts an oncotic pressure of 17.3 mmHg (ie 19.32 x 0.9). Measurement in an oncometer shows the actual plasma oncotic pressure is about 25 mmHg which is equivalent to a plasma protein concentration of 1.3 mmol/kg.

2.4.3: Measured versus predicted values

Why is there a difference between the actual measured value and the value calculated using the van't Hoff equation?

The two reasons are:

- The proteins are charged & non-permeant => Gibbs-Donnan effect
- The proteins are large => Excluded Volume effect

As protein is both charged and non-permeant across the capillary membrane, it sets up a Gibbs-Donnan equilibrium which affects the concentration of the diffusable ions on both sides of the membrane. The net result in this case is an increase in the number of particles per unit volume on the intravascular side of the membrane. The protein concentration appears to be larger than it is because of these extra particles and the effective oncotic pressure is therefore increased.

Additionally as protein molecules are large the volume they occupy in the solution is significant and this is another reason for the discrepancy from the van't Hoff equation which is derived for infinitely dilute (ie ideal) solutions. This second effect is known as the excluded volume effect.

What is the nature of these extra particles which are contributing an extra 7 to 8 mmHg (equivalent to about 0.4 mOsm/kg) to the measured oncotic pressure? They are mostly Na^+ ions as these are the cations present in by far the highest concentration. The $[Na^+]$ in plasma is increased by about 6 to 7 mmol/l because of the Gibbs-Donnan effect.

This increase of 6 to 7 mmol/l is much more than the 0.4 mOsm/l required to account for the increase in oncotic pressure. What is the explanation for this apparent discrepancy? The answer is that there is also a decrease in anion concentration which counteracts much of the increase in cation (ie Na^+) concentration so the *net* change in concentration is an increase of 0.4 mOsmoles/l.

Final note: Additional complicating factors not discussed here are that the net charge on the proteins is affected by:

- temperature
- pH
- types of protein present (eg albumin versus globulins)

CHAPTER OVERVIEW

3: Water Balance

3.1: Water Turnover

3.1.1: Water Balance

Water turnover is considered in terms of external balance and internal fluxes.

External balance refers to the comparison between the water input from and the water output to the external environment. Over any period of time, input equals output and the organism is in water balance.

Internal balance or flux refers to the movement of water across the capillaries of the body (including the secretion and absorption of the various transcellular fluids) and movement of water between interstitial and intracellular fluids.

3.1.2: External Balance

Total body water volume is tightly controlled with sensitive mechanisms that respond to changes in osmolality or intravascular volume. Estimates of daily water requirements are based several factors but that based on metabolic rate is probably the most accurate.

Estimation of Daily Water Requirements in Unstressed Healthy Adults

- Based on metabolic rate 80-110 mls/100kcals
- Based on body surface area 1.5 l/m2/day
- Based on weight 30-40 mls/kg/day

In the presence of disease, these estimates of fluid requirements are unreliable. Fluid administration should always be based on clinical circumstances (eg blood loss, internal fluid loss in third spaces, abnormal haemodynamics requiring intravascular volume loading, oliguria with acute renal failure). These disorders will not be discussed here as the emphasis is on normal physiology.

Water is required to replace losses which normally consist of insensible losses (from the skin and respiratory tract), urine, sweating and faecal loss.

An obligatory urine loss occurs because of the need to remove various solutes from the body. Other losses (eg sweating and faecal losses) are quite small under normal conditions. Faecal water loss averages about 200 mls/day but diarrhoea can be associated with large fluid & electrolyte losses.

Daily water requirements can vary greatly. The minimum water required for urine is dependent on the daily solute excretory load and the maximum urinary concentration achievable. For example, a typical daily solute load of 600 mOsms in a patient with a maximum urinary concentrating ability of 1200 mOsm/kg will require a minimum urine volume of 500mls/day to excrete it. If urine volume was less than this amount, solutes would accumulate and renal failure would be present. Ill or elderly patients are typically not able to achieve urine osmolality of 1200 mOsm/kg so the obligatory minimum urine volume required for solute excretion can be much higher than 500 mls.

The minimal amount of fluid loss from the body that can occur is referred to as the obligatory water loss. This sets a figure for the minimal amount of fluid intake that is required to maintain total water balance.

Components of Daily Obligatory Water Loss

- Insensible loss: 800 mls
- Minimal sweat loss: 100 mls
- Faecal loss: 200 mls
- Minimal urine volume to excrete solute load: 500 mls
- **Total: 1,600 mls**

The typical values in this example total to 1,600 mls. In stressed individuals, this obligatory loss may be much higher. Obligatory urine volume is variable (eg. the solute load may be decreased or the maximum urine concentration may be much lower then 1200 mOsm/l). There is also an inter-relationship between these two factors: as daily solute load increases, the maximum urine osmolality decreases until, at high solute loads, it is the same as plasma osmolality. This occurs because the increased urine flows necessary to carry the increased solute washes out the medullary osmoles and the time spent in the tubules is decreased.

Fluid input is from 2 major sources:

- **External**: Oral intake of fluids and food (and/or IV fluids)
- **Internal**: Metabolic water production

Food is an important source of water as nearly all food was once living (i.e. cellular) and has a high water content. Some processed foods may have a very low water content.

Metabolic water is water produced during the oxidation of food. Carbohydrates are completely metabolised to carbon dioxide and water. Metabolic water is about 350 to 400 mls/day (ie 5 mls/kg). This offsets some of the obligatory water losses.

Water intake in excess of requirements is excreted as urine. The other routes of fluid loss are not under regulatory control for maintaining water balance. The kidney is effectively the major effector organ in excreting excess water.

Hospitalised patients may have other sources of fluid input or loss (eg. IV fluids, vomiting, third space losses, diarrhoea) and the volumes involved can be very significant.

3.1.3: Internal Fluxes

The net movement of water between the intravascular and interstitial compartments across the capillary membrane depends on the balance of hydrostatic and oncotic pressures as described by Starling's hypothesis.

The major factor governing net movement of water between the ICF and the ISF is the osmotic forces.

Some examples of internal fluxes are:

- Diffusional turnover of water in the body's capillaries - this is huge: about 80,000 liters a day.
- Lymph flow is about 1 to 2.5 l/day with normal activity. [See Section 3.4: Lymph]
- Glomerular filtration rate (GFR) is 180 liters a day and the vast majority of this fluid is reabsorbed in the renal tubules.
- Turnover of fluids in the bowel is about 8 to 9 l/day. [See Section 3.5:Fluid and the bowel]

The diffusional turnover is very much higher than the net fluid loss from the capillaries via filtration.

3.2: Insensible Water Loss

What is 'insensible' water loss?

This term refers to water loss due to:

- Transepidermal diffusion: water that passes through the skin and is lost by evaporation, *and*
- Evaporative water loss from the respiratory tract

It is termed *insensible* as we are not aware of it.

KEY POINT: This is loss of pure water: there is no associated solute loss.

This solute-free water loss differs from sweating as sweat contains solutes. Insensible loss is different from sweating.

Insensible loss from the skin cannot be eliminated. Daily loss is about 400 mls in an adult.

Insensible loss from the respiratory tract is also about 400 mls/day in an unstressed adult. The water loss here is variable: it is increased if minute ventilation increases and can be decreased if inspired gas is fully humidified at a temperature of 37°C (e.g. as in a ventilated ICU patient).

The minimal insensible loss in an adult is about 800 mls/day. This is equivalent to a heat loss of about 480 kcals/day which is about 25% of basal heat production. On an average unstressed day, activity will increase insensible respiratory water losses so that the overall insensible loss is more than the minimum: an estimate of 50 mls/hr has been suggested for use in unstressed hospitalised patients. In clinical calculations of fluid balance, insensible losses are unmeasured are are usually accounted for by an estimate such as the one above. Metabolic water production (400 mls/day) is also unmeasured and can be considered to replace up to 50% of the insensible losses.

In simple bedside analysis of a patient's fluid balance, insensible loss is ignored as it cannot be measured. Similarly metabolic water production is ignored in a quantiative analysis of daily fluid balance. Even more significantly sweating is also ignored and volume loss with sweating can be quite large. The clinician doesn't have much choice as the volumes of these fluids cannot be routinely measured. The typical clinical practice is to calculate water input as (oral fluids + IV fluids) and water loss as (urine + other measured losses) and make a clinical estimate of the additional fluids required. The clinical estimate is based on factors such as assessment of the blood volume (BP including postural drop, pulse rate, CVP, urine flow rate, evidence of peripheral vasoconstriction based on colour and temperature) and knowledge of the pathophysiology of the disease process (such as typical expected losses e.g. in burn injury).

This is the best available approach because it is dependent mostly on clinical endpoints.

3.3: Sweat

Sweating is important for body temperature regulation but can also be a major source of water and solute loss. The heat loss can be quite significant because there is a loss of 0.58 kcals for every ml of water evaporated.

Maximum rate of sweating is up to 50 mls/min or 2,000 mls/hr in the acclimatised adult. This rate cannot be sustained but losses up to 25% of total body water are possible under severe stress: this could be fatal.

Losses due to Sweating

- Fluid loss - Can be large in a hot environment, or if physically active
- Solute loss - Decreases with 'acclimatisation'
- Heat loss - Can be large due to high latent heat of evaporisation of water; hence important role in body temperature regulation

There are several different types of sweating, but from the fluid perspective, only the sweating from eccrine sweat glands is important. The volume of fluid from apocrine sweating is very low.

The eccrine glands are specialised skin appendages which are present in over 99% of the skin surface. They are innervated by sympathetic cholinergic neurones. The muscarinic receptors can be blocked by atropine and this will prevent sweating.

Control

Sweating is controlled from a center in the preoptic & anterior regions of the hypothalamus where thermosensitive neurones are located. The heat regulatory function of the hypothalamus is also affected by inputs from temperature receptors in the skin. High skin temperature reduces the hypothalamic set point for sweating and increases the gain of the hypothalamic feedback system in response to variations in core temperature. Overall though, the sweating response to a rise in hypothalamic temperature (core temp) is much larger than the response to the same increase in average skin temperature.

Sweat is not pure water; it always contains a small amount (0.2 - 1%) of solute . When a person moves from a cold climate to a hot climate, adaptive changes occur in their sweating mechanism. These are referred to as acclimatisation: the maximum rate of sweating increases and its solute composition decreases. The daily water loss in sweat is very variable: from 100 to 8,000 mls/day. The solute loss can be as much as 350 mmols/day (or 90 mmols/day acclimatised) of sodium under the most extreme conditions. In a cool climate & in the absence of exercise, sodium loss can be very low (less than 5 mmols/day). $[Na^+]$ in sweat is 30-65 mmol/l depending on degree of acclimatisation.

Main Differences between Sweat and Insensible Water Loss

	Sweat	Insensible Fluid
Source	From specialised skin appendages called sweat glands	From skin (trans-epithelial) and respiratory tract
Solute loss	Yes, variable	None
Role	Body temperature regulation	* Cannot be prevented. * Evaporation of insensible fluid is a major source of heat loss from the body each day but is not under regulatory control

3.4: Lymph

3.4.1: What is Lymph?

Lymph is the name given to interstitial fluid which enters the lymphatic vessels.

Lymphatic capillaries are present in nearly all tissues. Significant exceptions are the central nervous system and bone. Small interstitial channels are present in the brain and the fluid flows into the CSF and then passes back into the circulation via the arachnoid villi.

The lymph capillaries are blind-ending and possess flap valves between adjacent lymphatic endothelial cells. These functional valves permit entry of ISF but prevent its return to the interstitium. The pressure inside the lymph capillary is about 1 mmHg at rest and the flap valves are closed. The lymph capillaries interconnect and join together to form lymph venules, and then large lymph veins which drain via lymph nodes into the thoracic duct (on the left) and the right lymphatic duct. By these two final pathways, lymph returns into the circulation.

Factors in Lymph Flow

- There is no central pump in the lymphatic system
- Forward flow is due to a pressure gradient within lymph vessels aided by one-way valves which prevent backflow
- Lymph enters lymph capillaries when the pressure in the tissue in low (up to 2mmHg) as the flap valves between lymph capillary cells are open
- ISF enters lymphatic capillaries in the phase after the external pressure has passed as external connective tissue fibres tend to tent open the lymph capillaries, opening the flap valves
- When ISF pressure increases beyond +2 mmHg then these flap valves close (passively due to the pressure gradient)
- With flap valves closed, the increased external (ISF) pressure tends to promote forward lymph flow provided pressure is not too high (eg ≤ 2 mmHg). At higher pressures, the unevenness of the pressure tends to close proximal lymph channels and lymph does not flow (Starling resistor effect)
- The main sources of suitable levels of external pressure to promote flow are arterial pulsations and muscular contractions
- The close association of lymph channels with arteries tends to favour flow
- Larger lymph vessels have smooth muscle in their walls. 'Intrinsic contraction' of these smooth muscle cells assists forward flow
- Lymph vessels have bi-leaflet valves every few mm and these are extremely important: no forward flow is ever lost

3.4.2: Functions of Lymph

The three functions of the lymphatic system are:

- Return of protein and fluid from the ISF to the circulation to maintain a low interstitial fluid protein concentration and maintain the oncotic pressure gradient across the capillary membrane. Oedema will occur if ISF oncotic pressure is not kept low.
- Role in absorption and transport of fat from the small intestine.
- Immunological role -lymph glands, and circulation of immune cells such as lymphocytes and dendritic cells, removal of bacteria.

Lymph from most parts of the body usually has a low protein concentration. Liver lymph is different because:

- It normally has a high protein concentration (due to low reflection coefficient)
- It contributes more than half of all the thoracic duct lymph

Consequently, the average lymph protein concentration in thoracic duct lymph is much higher than expected based on protein concentration in lymph from other body tissues.

The thoracic duct carries about 80% of the total lymph flow. This total flow at rest is about 120 mls/hr. If interstitial hydrostatic pressure rises (ie becomes less negative) due to excess fluid filtration & accumulation, the total lymph flow can increase quite markedly.

Chyle is lymph from the intestines which has a milky-while appearance due to the presence of large numbers of chylomicrons. Chylomicrons are 100nm diameter complexes of mostly triglycerides (containing the long chain fatty acids) enclosed in a hydrophobic protein coat. Chylomicrons enter the lymphatic lacteals in the villi, travel in the lymph and then enter the circulation via the thoracic duct.

Absorption of snake venoms (for Australian elapid snakes) occurs principally via lymph channels. If the bite is on a limb, the rate of venom absorption can be very much retarded by firm external compression of the lymph channels (pressure) and by not exercising the muscles of the limb (immobilisation). The aim of this 'pressure-immobilisation technique' for bites on limbs is to minimise entry of venom into the circulation and to 'buy time' so the person can reach medical care where specific anti-venom is available. As absorption is not directly into the venous system at the bite site, a torniquet is unnecessary and should NOT be used.

3.5: Fluid and the Bowel

The fluid in the bowel is generally considered as part of the transcellular fluid compartment. Turnover of fluid in the bowel is large; about 9 to 10 liters of fluid enter the gut each day:

Water Turnover in the Bowel

Water from diet	2000-3000 mls/day
Saliva	1000-2000
Gastric juice	1000-2000
Bile	500-1000
Pancreatic juice	1000-2000
Intestinal secretions	1000-2000

About 98% of this fluid is reabsorbed resulting in a faecal water loss of only 200 mls/day

This reabsorption occurs predominantly in the jejunum and ileum. About 1500 mls/day enter the colon from the ileum. This means that over a litre per day is absorbed in the colon.

The intestinal contents are essentially isotonic by the time the jejunum is reached because water can move into or out of the intestine in response to any osmotic gradient. Excess loss of intestinal content does not directly cause changes in osmolality of body fluids. As absorption of substances occurs, water moves along passively because of the osmotic gradient that is created. The colon is involved in reabsorbing water and electrolytes. Na^+ is actively reabsorbed and water again is reabsorbed passively down its osmotic gradient. Faecal loss of sodium is only about 5 mmol/day.

It is more accurate to consider the net fluid movements in the bowel as a cycling of fluid rather than a turnover of fluid. This cycling of fluid into the gut and back to the circulation each day has been called the enterosystemic circulation.

Bowel fluid loss may be internal or external. External losses include vomiting, diarrhoea and fistulae losses. Internal losses refer to sequestration of fluid into the bowel as part of the non-functional ECF or 'third space'. The direct result of these fluid losses is an isotonic contraction of the ECF. Electrolyte disturbances are common but vary depending on the condition and renal effects. Renal retention of water occurs with hypovolaemia and tends to cause hyponatraemia.

In small bowel obstruction, about 1500 mls of fluid is rapidly pooled in the bowel. By the time vomiting occurs, about 3000 mls of fluid is in the bowel. If the patient is hypotensive, then about 6000 mls is pooled in the intestines. Significant intravenous fluid resuscitation is usually required before operation in patients with bowel obstruction.

Apart from the gastric juice, all the other secretions into the bowel are alkaline with high [HCO_3^-]

Abnormal fluid losses from the bowel cause acid-base disturbances and these can be quite severe. The typical situation is:

- vomiting causes metabolic alkalosis (gastric alkalosis) with associated hyponatraemia, hypochloraemia & hypokalaemia
- acute diarrhoea (esp infective) causes a hyperchloraemic normal anion gap metabolic acidosis
- chronic diarrhoea (esp non-infective) can cause a metabolic alkalosis.

In summary:

- A large amount of fluid is cycled through the bowel each day. (net reabsorption = 98%) Intestinal fluids are isotonic
- Water is reabsorbed passively down its osmotic gradient created by the active reabsorption of nutrients and electrolytes. Electrolyte & acid-base disturbances can occur with abnormal bowel fluid losses.

3.6: Cerebrospinal Fluid

Cerebrospinal fluid (CSF) is considered a part of the transcellular fluids. It is contained in the ventricles and the subarachnoid space and bathes the brain and spinal cord. The CSF is contained within the meninges and acts as a cushion to protect the brain from injury with position or movement. It has been estimated that this cushioning or 'water bath' effect gives the 1400g brain an effective net weight of only 50g.

The total volume of CSF is 150 mls. The daily production is 550 mls/day so the CSF turns over about 3 to 4 times per day. The CSF is formed by the choroid plexus (50%) and directly from the walls of the ventricules (50%). CSF flowsthrough the foramens of Magendie & Luschka into the subarachnoid space of the brain and spinal cord. It is absorbed by the arachnoid villi (90%) and directly into cerebral venules (10%).

The normal intracerebral pressure (ICP) is 5 to 15 mmHg. The rate of formation of CSF is constant and is not affected by ICP. Absorption of CSF increases linearly as pressure rises above about 7 cmsH_2O pressure. At a pressure of about 11cms H_2O, the rate of secretion & absorption are equal.

The CSF has a composition identical to that of the brain ECF but this is different from plasma. The major differences from plasma are:

- The pCO_2 is higher (50 mmHg) resulting in a lower CSF pH (7.33)
- The protein content is normally very low (0.2g/l) resulting in a low buffering capacity
- The glucose concentration is lower
- The chloride concentration is higher
- The cholesterol content is very low

There are no lymphatic channels in the brain and CSF fulfils the role of returning interstitial fluid and protein to the circulation.

The CSF is separated from blood by the blood-brain barrier. Only lipid soluble substances can easily cross this barrier and this is important in maintaining the compositional differences.

CHAPTER OVERVIEW

4: Capillary Fluid Dynamics

4.1: Microcirculation

What is the 'microcirculation'?

The microcirculation refers to the smallest blood vessels in the body:

- the smallest arterioles
- the metarterioles
- the precapillary sphincters
- the capillaries
- the small venules

The lymph vessels are not included. The arterioles contain vascular smooth muscle and are the major site of systemic vascular resistance. In skeletal muscle and other tissues, a large number of capillaries remain closed for long periods due to contraction of the precapillary sphincter. These capillaries provide a reserve flow capacity and can open quickly in response to local conditions such as a fall in pO_2 when additional flow is required.

The microcirculation of some tissues (eg skin) contain direct AV connections which act as shunts. The flow in these shunts does not participate in transfer of gases, nutrients or wastes. These AV shunts are under the control of the nervous system. In the skin, opening or closing of these shunts is important in heat regulation.

The smooth muscle of the metarterioles and the precapillary sphincters contracts and relaxes regularly causing intermittent flow in the capillaries: this is known as vasomotion. A local drop in pO_2 is the most important factor causing relaxation of the precapillary sphincters. The intermittent flow is not due to the cyclical rise and fall of the blood pressure as these fluctuations are almost completely damped out by the arterioles.

The principal function of the microcirculation is to permit the transfer of substances between the tissues and the circulation. This transfer occurs predominantly across the walls of the capillaries but some exchange occurs in the small venules also. Substances involved include water, electrolytes, gases (O_2, CO_2), nitrogenous wastes, glucose, lipids and drugs.

Electrolytes and other small molecules cross the membrane through pores. Lipid soluble substances (including oxygen and carbon dioxide) can also easily cross the thin (1 mm) capillary walls. Proteins are large and do not cross easily via pores but some transfer does occur via pinocytosis (endocytosis/exocytosis).

Water molecules are smaller than the size of the pores in the capillary and can cross the capillary wall very easily. The capillary endothelial cells in some tissues (eg glomerulus, intestinal mucosa) have gaps (called fenestrations) in their cytoplasm which are quite large. The water conductivity across these capillaries is much higher then in non-fenestrated capillaries in other tissues of the body.

The transfer of water across the capillary membrane occurs by two processes: diffusion and filtration.

Diffusion

The total daily diffusional turnover of water across all the capillaries in the body is huge (eg 80,000 liters per day) and is much larger than the total capillary blood flow (cardiac output) of about 8,000 liters per day.

Diffusion occurs in both directions and does not result in net water movement across the capillary wall. This is because net diffusion is dependent on the presence of a concentration gradient for the substance (Fick's Law of Diffusion) and there is ordinarily no water concentration difference across the capillary membrane. Net diffusional flux is zero.

Filtration

This is actually ultrafiltration as the plasma proteins do not cross the capillary membrane in most tissues. This filtration is considered to occur because of the imbalance of hydrostatic pressures and oncotic pressures across & along the capillary membrane (Starling's hypothesis - see following section).

For the whole body, there is an ultrafiltration outward of 20 liters per day and inwards of 18 liters per day. The difference (about 2 liters/day) is returned to the circulation as lymph.

Filtration results in net movement of water because there is an imbalance between the forces promoting outward flow and the forces promoting inward flow. These forces are variable so net movement could be inwards or outwards in a particular tissue at a certain time. The forces also change in value along the length of the capillary and the typical situation is to have net movement outward at the arterial end and to have net movement inward at the venous end of the capillary.

Comparison of Diffusion and Filtration in Capillaries

Diffusion:

- HUGE volumes of water are involved
- bidirectional along the whole length of the capillary
- net movement is governed by concentration gradient
- hydrostatic & oncotic pressures (Starling forces) are not involved in diffusion
- this is the process responsible for net movement of gases, nutrients and wastes (as these substances move down their concentration gradients)
- there is no net movement of water across the capillary wall due to diffusion -this is remarkable considering the huge volume of water involved.

Filtration

- really ultrafiltration as proteins cannot easily cross most capillary membranes
- volumes involved are much smaller then the diffusional flux
- fluid movement can be either inwards (absorption) or outwards, but not both at any particular position along the capillary
- net movement is governed by the balance of the hydrostatic and oncotic pressure gradients (the Starling forces)
- this process is not important for net movement of gases, nutrients & wastes the net movement of water is important.

Fick's Law of Diffusion

This states that the amount of diffusion (or flux) of a substance across any membrane is proportional to the concentration difference (C2-C1) across the membrane and to the surface area (A) of the membrane and is inversely proportional to the thickness (t) of the membrane. The constant of proportionality (k) is a measure of the permeability of the membrane to the substance:

$$Flux = k \times \frac{A \cdot (C2 - C1)}{t}$$

4.2: Starling's Hypothesis

Starling Forces and Factors

A quote from Starling (1896)

"... there must be a balance between the hydrostatic pressure of the blood in the capillaries and the osmotic attraction of the blood for the surrounding fluids. "

" ... and whereas capillary pressure determines transudation, the osmotic pressure of the proteids of the serum determines absorption."

Starling's hypothesis states that the fluid movement due to filtration across the wall of a capillary is dependent on the balance between the hydrostatic pressure gradient and the oncotic pressure gradient across the capillary.

The four Starling's forces are:

- hydrostatic pressure in the capillary (Pc)
- hydrostatic pressure in the interstitium (Pi)
- oncotic pressure in the capillary (pc)
- oncotic pressure in the interstitium (pi)

The balance of these forces allows calculation of the net driving pressure for filtration.

$$\text{Net Driving Pressure} = [(Pc - Pi) - (pc - pi)]$$

Net fluid flux is proportional to this net driving pressure. In order to derive an equation to measure this fluid flux several additional factors need to be considered:

- the reflection coefficient
- the filtration coefficient (Kf)

An additional point to note here is that the capillary hydrostatic pressure falls along the capillary from the arteriolar to the venous end and the driving pressure will decrease (& typically becomes negative) along the length of the capillary. The other Starling forces remain constant along the capillary.

The reflection coefficient can be thought of as a correction factor which is applied to the measured oncotic pressure gradient across the capillary wall. Consider the following:

The small leakage of proteins across the capillary membrane has two important effects:

- the interstitial fluid oncotic pressure is higher then it would otherwise be.
- not all of the protein present is effective in retaining water so the effective capillary oncotic pressure is lower than the measured oncotic pressure (in the same way that there is a difference between osmolality and tonicity).

Both these effects decrease the oncotic pressure gradient. The interstitial oncotic pressure is accounted for as its value is included in the calculation of the gradient.

The reflection coefficient is used to correct the magnitude of the measured gradient to take account of the effective oncotic pressure. It can have a value from 0 to 1. For example, CSF & the glomerular filtrate have very low protein concentrations and the reflection coefficient for protein in these capillaries is close to 1. Proteins cross the walls of the hepatic sinusoids relatively easily and the protein concentration of hepatic lymph is very high. The reflection coefficient for protein in the sinusoids is low. The reflection coefficient in the pulmonary capillaries is intermediate in value: about 0.5.

Starling Equation

The net fluid flux (due to filtration) across the capillary wall is proportional to the net driving pressure. The filtration coefficient (Kf) is the constant of proportionality in the flux equation which is known as the Starling's equation.

$$J_v = L_p S([P_c - P_i] - \sigma[\pi_p - \pi_i]$$

The **filtration coefficient** consists of two components as the net fluid flux is dependent on:

- the area of the capillary walls where the transfer occurs
- the permeability of the capillary wall to water. (This permeability factor is usually considered in terms of the hydraulic conductivity of the wall.)

The filtration coefficient is the product of these two components: $Kf = Area \times \text{Hydraulic conductivity}$

A leaky capillary (eg due to histamine) would have a high filtration coefficient. The glomerular capillaries are naturally very leaky as this is necessary for their function; they have a high filtration coefficient

Typical values of Starling Forces in Systemic Capillaries (mmHg)

	Arteriolar end of capillary	**Venous end of capillary**
Capillary hydrostatic pressure	25	10
Interstitial hydrostatic pressure	-6	-6
Capillary oncotic pressure	25	25
Interstitial oncotic pressure	5	5

The net driving pressure is outward at the arteriolar end and inward at the venous end of the capillary. This change in net driving pressure is due to the decrease in the capillary hydrostatic pressure along the length of the capillary.

The values quoted in various sources vary but most authors adjust the values to ensure the net gradients are in the appropriate direction they wish to show. The method (used in some sources) of just summing the various forces takes no account of the reflection coefficient. The values for hydrostatic pressure are not fixed and vary quite widely in different tissues and indeed within the same tissue. Contraction of precapillary sphincters and/or arterioles can drop the capillary hydrostatic pressure quite low and the capillary will close.

When first measured by Landis in 1930 in a capillary loop in a finger held at heart level, the hydrostatic pressures found were 32 mmHg at the arteriolar end and 12 mmHg at the venous end. The later discovery of negative values for interstitial hydrostatic pressure by Guyton did upset the status quo a bit.

The Starling equation cannot be used quantitatively in clinical work. To actually use the Starling equation clinically requires measurement of six unknowns. This is simply not possible and this limits the usefulness of the equation in patient care. It can be used in a general way to explain observations (eg to explain generalised oedema as due to hypoalbuminaemia).

Special Cases of Starling's Equation

The microcirculations of the kidney, the lung and the brain are special cases in the use of the Starling equation and are considered in the next three sections.

4.3: Ultrafiltration in the Glomerulus

The situation in the glomerular capillaries is quite remarkable. In the rest of the body, the net excess of ultrafiltration over reabsorption is of the order of two to four liters a day. The net excess in the glomerular capillaries is known as the glomerular filtration rate (GFR) and is 180 litres/day.

The situation in the glomerulus

The filtration coefficient is high (mostly because of a high permeability but also because of a large surface area)
The reflection coefficient is high: about 1.0 (i.e. the filtrate is a true ultrafiltrate as the glomerular capillaries are essentially impermeable to protein (so oncotic pressure in the filtrate is zero)
The hydrostatic pressure in the capillaries is high and does not decrease much along the length of the capillary
Because of the large loss of fluid and the impermeability to protein, the oncotic pressure in the glomerular capillary increases along its length. (This increased oncotic pressure in important in the reabsorption from the proximal tubule into the peritubular capillaries)
There is a net outward filtration pressure often along the whole length of the capillary.

Typical values of Starling Forces in Glomerular Capillaries (mmHg)

	Aff. Arteriolar End	***Eff. Arteriolar End***
Hydrostatic pressure in capillary (HP_{GC})	60	58
Hydrostatic pressure in Bowman's capsule (HP_{BC})	15	15
Oncotic pressure in capillary (OP_{GC})	21	33
Oncotic pressure in Bowman's capsule (OP_{BC})	0	0
Net Filtration Pressure	*24*	*10*

The flux equation discussed earlier simplifies to just 4 terms:

$$GFR = K_f \times (HP_{GC} - HP_{BC} - OP - GC)$$

The term that varies along the length of the glomerular capillary is OP_{GC}. This is a quite a different situation to what occurs in most tissue capillary beds, where the change that occurs along the length of the capillary is a decrease in capillary hydrostatic pressure. The glomerulus is different because of the very large fluid loss that occurs.

The hydrostatic pressure in the glomerular capillaries is affected by the balance between afferent and efferent arteriolar constriction.

4.4: Pulmonary Microcirculation

Gas exchange is the prime function of the lung. The pulmonary circulation moves the pulmonary blood into close association with the alveoli (at the blood-gas barrier) so that gas exchange is facilitated. The flow involved is large as the pulmonary blood flow is equal to the cardiac output. Efficient gas exchange is facilitated because the blood-gas membrane is thin with a large surface area. At any moment, the pulmonary capillary blood volume is about 80 mls.

The key features of the pulmonary microcirculation are:

- The pulmonary capillaries (and the alveoli) have very thin walls which minimises the barrier to diffusion.
- In the alveolar walls, the capillaries form a dense network which has been considered to be almost a continuous thin film of blood. This provides a large capillary surface area.
- The pressures in the pulmonary circuit are much lower than in the systemic circulation and the pulmonary vascular resistance is very low. The pressure is just sufficient to perfuse the apical areas of the lungs in the erect healthy adult.

The Starling equation can be applied to the pulmonary microcirculation in the same way as any other capillary bed.

Typical values for the Starling's Forces in Pulmonary Capillaries

Capillary hydrostatic pressure (Pc) is 13 mmHg (arteriolar end) to 6 mmHg (venous end) but variable because of the hydrostatic effects of gravity esp in the erect lung.
Interstitial hydrostatic pressure (Pi) - Variable but ranges from zero to slightly negative.
Capillary oncotic pressure = 25 mmHg (Same as in systemic capillaries)
Interstitial oncotic pressure = 17 mmHg (This is estimated from measurements on lung lymph)

Oncotic pressure gradient

The interstitial oncotic pressure is high indicating significant leak of protein (mostly albumin) across the thin capillary walls under normal circumstances. The reflection coefficient has been estimated at about 0.5

Considering the typical values and allowing for the reflection coefficient, it can be estimated that the net oncotic gradient is small but favours reabsorption.

Hydrostatic pressure gradient

The capillaries are called intra-alveolar vessels and the presssure they are exposed to is close to alveolar pressure (which has an average value of zero). However, actual measurements of pressure in the alveolar interstitium have found slightly negative pressures (eg -2 mmHg). Closer to the hilum, the interstitial pressures become more negative and this favours flow of fluid from the alveolar intersitium into the pulmonary lymphatics.

The capillary hydrostatic pressure is variable because of the effects of gravity. Consider: The erect lung is basically suspended in a gravitational field so the pressure in the vessels at the base of the lung is higher than the pressure at the apex. The pressure difference is equivalent to the height of a static water column from the base to the apex. The distance involved is about 30 cms so the pressure difference is 30 cms H_2O which is about 23 mmHg. If the typical pulmonary artery pressure is 25/8 then it is apparent that the pressure is just adequate for perfusion of the apex of the erect lung.

The pulmonary circuit has a low resistance and about half of this resistance is due to the pulmonary capillaries which have no muscle in their walls. The capillary hydrostatic pressure is quickly affected by changes in pulmonary artery pressure and left atrial pressure without much protective buffering.

Overall Effect

The balance of Starling forces in the lung is generally stated as favouring reabsorption because of the clinical fact that the lungs are generally dry and clearly need to be to facilitate gas exchange. Under normal conditions, there is a small net outward movement of fluid. This is estimated as equal to the pulmonary lymph flow rate. The flow is usually small (eg 10 to 20 mls/min) which is only about 2% of the pulmonary blood flow. So despite the net outward hydrostatic pressure gradient and the high reflection coefficient which limits the effectiveness of the oncotic pressure in opposing outward fluid movement, the measured low lymph flow means that the balance of forces is clearly to minimise loss of fluid into the interstitium.

The large surface area and thin capillary walls which assist efficient gas exchange also facilitate filtration from the capillaries to the interstitium. The interstitial fluid move towards the hilum along the spaces beside the vessels and the airways. The interstitial hydrostatic pressure probably becomes more negative as the hilum is approached. The excess filtrate is removed by the pulmonary lymphatics. Lymphatic flow is promoted by the rhythmic external compression that occurs during the ventilatory cycle and by the presence of one way valves.

The Starling equation is not very useful clinically because it is not possible to measure all six of the unknown values. In particular, bedside determination of the interstitial hydrostatic & oncotic pressures and the reflection coefficient is not possible. The clinician is limited to assessments based on plasma protein concentration (as index of capillary oncotic pressure) and values obtainable from use of a pulmonary artery catheter (eg wedge pressure as estimate of left atrial pressure & mean pulmonary venous pressure). A clinical examination and a chest xray are much more useful in assessing & monitoring pulmonary oedema.

Safety Factors Preventing Pulmonary Oedema

For pulmonary oedema to occur, excess fluid must first accumulate in the interstitium (interstitial oedema), then must move into the alveoli (alveolar flooding). The lung is relatively resistant to the onset of pulmonary oedema and this is usually ascribed to several safety factors:

- Increased lymph flow: Increased fluid filtration causes increased lymph flow which tends to remove the fluid.
- Decrease in interstitial oncotic pressure (oncotic buffering mechanism): When filtration increases, the albumin loss in the filtrate decreases. This combined with the increased lymph flow washes the albumin out of the interstitium and interstitial oncotic pressure decreases. This protection does not work if the capillary membrane is damaged eg by septic mediators.
- High interstitial compliance: A large volume of fluid can accumulate in the gel of the interstitium without much pressure rise. Finally, the interstial tissues become full of fluid, the pressure rises and alveolar flooding occurs. This has been called the bathtub effect: the analogy is that the tub can take a lot of fluid but there comes a point when it is full and suddenly overflows.

These safety mechanisms are quite effective especially in preventing pulmonary oedema associated with rises in capillary hydrostatic pressure. It has been estimated that the capillary hydrostatic pressure can rise to three times normal before alveolar flooding occurs. Surfactant assists in the prevention of alveolar flooding also.

4.5: Blood-Brain Barrier

When considering the Starling hypothesis it is usual to consider the important special cases of the glomerulus and the lung. However, the situation with the cerebral capillaries is very different and this seems to be rarely appreciated. The capillary membranes in most of the body are permeable to the low molecular solutes present in blood but are more or less impermeable to the large molecular weight proteins. The only solutes present that can exert an osmotic force across the capillary wall in most of the capillaries are the proteins so the oncotic pressures of plasma and the interstitium are two important Starling's forces. The low molecular weight solutes can easily cross most capillary membranes so they are not effective at exerting an osmotic force across the capillary endothelial cells.

How are the brain capillaries different?

The differences are due to the blood-brain barrier:

The capillary membrane in the cerebral capillaries is relatively impermeable to most of the low molecular weight solutes present in blood (as well as to the plasma proteins).

The ions Na+ and Cl- make up most of these solutes. These solutes are effective at exerting an osmotic force across the cerebral capillary membrane (the site of the blood-brain barrier). As a consequence, the Starling's forces in the cerebral capillaries are:

- the hydrostatic pressure in the cerebral capillaries
- the hydrostatic pressure in the brain ECF (ICP)
- the osmotic pressure of the plasma
- the osmotic pressure of the brain ECF

Note that it is total osmotic pressure rather than oncotic pressure. The oncotic pressure is extremely small in comparison to the huge osmotic pressure exerted by the effective small solutes in the cerebral capillaries. The small leak of these low molecular weight solutes can be accounted for by a reflection coefficient as with the plasma proteins in other capillary beds. A one milliOsmole /kg increase in osmotic gradient between blood and brain interstitial fluid will exert a force of 17 to 20 mmHg. At an osmolality of 287 mOsm/kg then the total osmotic pressure is about 5400mmHg as can be calculated with the van't Hoff equation. In comparison, the plasma oncotic pressure of 25 mmHg is tiny.

Therefore even small changes in plasma tonicity can have a marked effect on the total fluid volume of the intracranial compartment. It is not just the intracellular volume of the brain cells but also the volume of the brain ECF that are decreased by an increase in plasma osmolality. In other tissues of the body, an increase in plasma osmolality would increase ISF volume but decrease ICF volume in that tissue.

Effect of Increase in Plasma Osmolality on Tissue Fluid Volumes

	ISF volume	ICF volume	Total fluid volume
Brain	Decreased	Decreased	ALWAYS decreased
Other tissues	Increased	Decreased	Dept on balance between the increased ISF & the decreased ICF

Infusion of hypertonic solutions of any effective small molecular weight solute (eg hypertonic saline, mannitol or urea) will dehydrate the brain. In the peripheral capillaries, these solutes are not effective at exerting an osmotic force because they can easily cross these capillary membranes. Hypertonic solutions of sodium (as saline) or mannitol are however effective at the cell membrane and will cause cellular dehydration in all body cells. Urea can cross most cell membranes relatively easily and is a much less effective solute at this membrane.

A final comment should also be made about the water permeability of the blood brain barrier. The fluid flux across the capillary membrane is proportional the the net pressure gradient (as stated in the Starling equation). The constant of proportionality in this equation is the filtration coefficient and the value of this is a measure of how easily water crosses the membrane. As discussed earlier, this filtration coefficient is the product of the total area of the capillary walls and the hydraulic conductivity. This hydraulic conductivity is a measure of the water permeability of the membrane. The point to make is that in comparison to other body capillaries the hydraulic conductivity (ie water permeability) of the cerebral capillaries is very much lower. This greatly minimises the amount of water that is lost from the brain in response to changes in plasma tonicity and this is fortunate in view of the huge changes in osmotic forces that occur with tonicity changes of only a few millOsmoles/kg. This very low filtration coefficient is necessary for maintaining a constant intracranial volume.

Note the difference between the reflection coefficient and the filtration coefficient

The reflection coefficient gives a measure of how well solutes cross a membrane and the filtration coefficient (or more accurately the hydraulic conductivity) gives a measure of how well the solvent (water) crosses a membrane. This distinction is important to consider in the brain because cerebral damage does not necessarily result in equal changes in each coefficient in the area of damage. For example it is often said that hypertonic mannitol solutions are less effective at dehydrating abnormal or damaged areas of the brain (as compared to normal areas) but this is not necessarily correct. A damaged area may have a lower reflection coefficient for low molecular weight solutes so an increase in osmotic gradient due to mannitol will be less effective in this area. However, the damaged area may also have a higher hydraulic conductivity and water is more able to leave the brain in this area. The net effect is that the damaged brain may be dehydrated as much as (or more) than undamaged areas.

Summary

The blood brain barrier is impermeable to low molecular weight solutes so the plasma osmotic pressure (rather than plasma oncotic pressure) is the Starling force to be considered here. For the same reason, the brain interstitial osmotic pressure is also a Starling force (rather then the oncotic pressure of the interstitial fluid).

The reflection coefficient due to these solutes is used rather than the reflection coefficient for the proteins. This reflection is very high for most of these water-soluble solutes.

The Starling equation is also altered for another reason: the hydraulic conductivity of the cerebral capillaries is very much lower than in other capillaries. The filtration coefficient is low. This minimises the amount of dehydration that occurs in response to changes in plasma tonicity. The application of the Starling equation to the brain is different from that anywhere else in the body and it is surprising this is so little appreciated especially in view of the important clinical relevance (eg use of hypertonic mannitol solutions).

Finally, because of Pascal's principle, the interstitial fluid pressure in the brain is equal to the CSF pressure (ie intracranial pressure).

The cerebral capillaries are indeed an important 'special case' as regards the application of Starling's hypothesis.

CHAPTER OVERVIEW

5: Control of Water Metabolism

5.1: Summary of Renal Water Handling

The kidney is the most important organ in the regulation of water balance in the body. Under normal circumstances, the sensitive hypothalamic osmoreceptors detect any change in extracellular tonicity and respond by altering secretion of ADH from the posterior pituitary. The volume receptors are much less sensitive and really function as a backup sensor. Most water intake is not due to thirst. The thirst mechanism functions as a backup effector mechanism.

The kidney is the effector organ for body water balance.

Glomerular filtration rate (GFR) is very large (180 l/day) in comparison to the amount of urine that is typically produced. Most of the water in the filtrate is reabsorbed because of renal processes which are independent of ADH action.

The diagram below summarises the percentages of water reabsorbed in the various renal segments. The two extreme examples of absence of ADH and maximal ADH production are outlined. The kidney adjusts the water reabsorption between these two extremes (under the influence of ADH) in order to maintain a constant plasma osmolality. The minimum (or obligatory) urine volume is determined by the size of the daily solute load and the maximal urine osmolality that can be achieved. The maximum urine osmolality decreases with increasing age in adults so the obligatory urine volume is higher for a given solute load in the elderly.

All water reabsorption in the kidney is passive.

Water moves in response to osmotic gradients. These osmotic gradients are all directly or indirectly due to the reabsorption of solute particularly sodium. There are no water pumps in the body.

Quantitative Summary of Renal Water Handling

GFR (180 l/day) 65% Reabsorbed Proximal Tubule 15% Reabsorbed Loop of Henle (thin descending limb) 20% of filtrate Enters Distal Tubule In absence of ADH With maximal ADH $8\%\text{Reab in CD} > 19\%\text{Reab in CD}$ & & $12\%\text{of filtered } H_2O < 1\% of filtered H_2O$ in urine in urine (22 l/day at 30-60 mOsm/l) (500 mls at 1200 mOsm/day)	to be completed

5.2: Overview

Elements of a Simple Control System

A basic control system for a regulated physiological variable consists of several components:

- Sensors -these are receptors which respond either directly or indirectly to a change in the controlled variable
- Central controller -this is the coordinating and integrating component which assesses input from the sensors and initiates a response
- Effectors -these are the components which attempt, directly or indirectly to change the value of the variable.

For a control system to function effectively, there must be a closed loop. The change due to the action of the effectors must be detected by the sensors. This monitoring by the sensors provides feedback to the central controller. This type of system is referred to as a servo-control system.

[Diagram to come]

For many apparently regulated physiological variables, it can be difficult to see how this control model fits. In some cases this is due to the complexity of many interacting factors and interacting control systems which are difficult to separate and state simply. However, a simple model for the control of water balance is easy to construct within this framework. It should be noted that additional mechanisms (eg local renal factors) can affect water balance quite significantly in some circumstances - some of these are discussed in Section 5.9.

Under normal circumstances, most water input is due to ingested water (as fluids or in food). The sensitive osmoreceptors adjust water balance by ADH-mediated changes in free water excretion into the urine and thirst-mediated changes in water ingestion. The mechanism for regulation of water balance is often referred to as the thirst-ADH mechanism. The following sections discuss the components of the control system in more detail.

5.3: Sensors for Control of Water Balance

The main sensors that are involved in control of water balance in the body are:

- Osmoreceptors
- Volume receptors (low pressure baroreceptors)
- High pressure baroreceptors

5.2.1: Osmoreceptors

The osmoreceptors are specialised cells in the hypothalamus which respond to changes in extracellular tonicity (rather then to changes in osmolality). The exact mechanism involved is not known but it may be that changes in cell volume affect the concentration of certain critical intracellular molecules or affect the activity of ion channels in the cell membrane.

As Na^+ (and its obligatory associated anions - mostly Cl^-, HCO_3^- & protein-) account for 92% of ECF tonicity, these receptors (during normal physiology) function essentially as monitors of ECF [Na^+]. These receptors have been called osmo-sodium receptors. This is not strictly correct as the variable directly sensed is tonicity and this can change independently of [Na^+] in certain non-physiological situations (eg mannitol infusion); but in nearly all physiological circumstances it is a functionally accurate statement.

Osmoreceptors are very sensitive

They respond to a change as small as a 1 to 2% increase in tonicity. Water intake can vary greatly but plasma osmolality varies only one to two percent because of the efficient and powerful control system coupled to these osmoreceptors.

These receptors are monitoring 'water balance' indirectly because they look at the effect of an excess or deficit of water by its effect on tonicity. This could cause a problem, if for example, both ECF water and solute increased together so that [Na^+] and tonicity remained constant. This is what happens with an intravenous infusion of normal saline (ie an isotonic expansion of the ECF). Fortunately the body has several mechanisms of recognising changes in intravascular volume. This is the role of the baroreceptors.

Note that the osmoreceptors effectively respond to the ECF [Na^+] and this is also the factor which effectively controls the distribution of water between intracellular and extracellular fluid. (See Section 6.1) The ECF [Na^+] thus sets the ECF volume and controls the ICF:ECF distribution of body water so it necessarily follows that:

ECF [Na^+] is an effective monitor of total body water

5.2.2: Baroreceptors

Effective intravascular volume can be independently assessed by the low pressure baroreceptors (volume receptors) which also provide input to the hypothalamus. These volume receptors are located in the right atria and great veins and respond to the transmural pressure in the the walls of these vessels.

Baroreceptors are less sensitive (but more potent) than the osmoreceptors.

The threshold of the volume receptors for causing changes in ADH secretion is an 8 to 10% change in blood volume. But when stimulated they cause ADH levels to be much higher than that seem with osmoreceptor stimulation.

Hypovolaemia is a more potent stimulus for ADH release than is hyperosmolality. A hypovolaemic stimulus to ADH secretion will override a hypotonic inhibition and volume will be conserved at the expense of tonicity. The maximum levels of ADH reached with a significant (20%) volume depletion is about 40pg/ml which is larger than the 12-15pg/ml reached with a maximum isovolaemic increase in osmolality.

The high pressure baroreceptors input to the hypothalamus via adrenergic pathways. These baroreceptors are located in the carotid sinus and respond to changes in mean arterial blood pressure. The input to the hypothalamus from the volume receptors and the high pressure baroreceptors rarely conflicts as hypovolaemia tends to be associated with hypotension (and vice versa).

5.2.3: Other Non-osmotic Stimuli

In addition to changes in intravascular volume, there are several other non-osmotic factors which affect ADH secretion. These include input from higher cerebral centres and various drugs.

5.4: The Central Controller in Water Balance

The central controller for water balance is the hypothalamus but there is no single anatomically defined center which is solely responsible for producing an integrated response to changes in water balance. There are numerous pathways interconnecting the various centers or areas in the hypothalamus. The osmoreceptors are located in the area known as the AV3V (anteroventral 3rd ventricle). Lesions in the AV3V region in rats cause acute adipsia.

The thirst centre is located in the lateral hypothalamus. It receives input from osmoreceptors in the AV3V region and from the subfornical organ and the organum vasculosum of the lamina terminalis (OVLT) which are sites for angiotensin II action. The OVLT is in the AV3V region.

ADH is formed predominantly in the neurones of the supraoptic and paraventricular nuclei. These nuclei receive input from the osmoreceptors and also from ascending adrenergic pathways from the low and the high pressure baroreceptors. Aquaporin 4 has recently been identified in cells in the hypothalamus particularly in the paraventricular and supraoptic nuclei.

The key parts of the hypothalamus involved in water balance are:

- Osmoreceptors
- Thirst centre
- OVLT & SFO (respond to angiotensin II)
- Supraoptic & paraventricular nuclei (for ADH synthesis)

5.5: Effector Mechanisms

The major effector mechanisms are:

- Thirst
- ADH (anti-diuretic hormone).

Control of Water Input : Thirst

Thirst is a mechanism for adjusting water input via the GIT.

Control of Water Output : ADH & the Kidney

ADH provides a mechanism for adjusting water output via the kidney. Note that ADH is often called 'vasopressin' - this term refers to the vasoconstrictive properties of very large doses ('pharmacological doses') of the hormone

Both thirst and ADH can increase when water is needed by the body and the usual physiological outcome is to repair the water deficit. These effector mechanisms are discussed in the next 2 sections. The whole system for control of water balance as outlined in this chapter is frequently referred to as "the thirst-ADH mechanism" though this really only refers to the effector part of the control system.

5.6: Thirst

What is thirst?

Thirst is "the physiological urge to drink water". In studies, it is recognised when subjects report the conscious sensation of a desire to drink. Under normal conditions, most water intake is due not to thirst but to social and cultural factors (eg drinking with meals or at work breaks, water in food). Thirst offers a backup to these behavioural factors and to the ADH response. Both the thirst and the ADH mechanisms are regulated in the hypothalamus. Water intake can be considered to consist of two components: a regulatory component (due to thirst) and a non-regulatory component (all other fluid intake).

Stimuli to Thirst

The 4 major stimuli to thirst are:

- Hypertonicity: Cellular dehydration acts via an osmoreceptor mechanism in the hypothalamus
- Hypovolaemia: Low volume is sensed via the low pressure baroreceptorsin the great veins and right atrium
- Hypotension: The high pressure baroreceptors in carotid sinus & aorta provide the sensors for this input
- Angiotensin II: This is produced consequent to the release of renin by the kidney (eg in response to renal hypotension)

There is strong evidence for a role of the octapeptide angiotensin II in physiological thirst: it is a potent dipsogen. The action is mediated via the effect of angiotensin II on specific receptors located in the subfornical organ (SFO) and the organum vasculosum of the lamina terminalis (OVLT). Both the SFO and the OVLT are circumventricular organs: they lie outside the blood-brain barrier allowing blood-borne substances (angiotensin II in this case) to affect neurones. The neuronal pathway from the SFO to the hypothalamus uses angiotensin II as a neurotransmitter. Ascending neural pathways arising from the low and high pressure baroreceptors enter the same area of the hypothalamus. Hypovolaemia and hypotension are facilitators for the development of thirst.

It is not known whether the osmoreceptor which stimulates thirst is the same or different from the one stimulating ADH release but they are located in the same area of the hypothalamus. The osmotic threshold for thirst may be set higher than that for ADH release but this is disputed. If it was, it would suggest that thirst has a backup role for situations where alterations in plasma tonicity are not corrected solely by ADH changes. Thirst and ADH release are interrelated in the hypothalamus via neuronal connections between relevant areas.

Outcome

Thirst leads to drinking. This is a powerful defence against hyperosmolality. As long as access to water is unrestricted and the person is able to drink, then significant hyperosmolality will not develop. For example, elderly patients with non-ketotic hyperglycaemia do not become significantly hyperosmolar unless water intake becomes restricted for some reason.

Drinking stimulates mechanoreceptors in the mouth and pharynx. These peripheral receptors provide input to the hypothalamus and the sensation of thirst is attenuated. This occurs even before any reduction in plasma tonicity. This may be a safeguard against over-ingestion of water as there is an inevitable delay before the ingested water is absorbed and available to decrease plasma osmolality.

5.7: Antidiuretic Hormone

5.6.1: ADH in the Hypothalamus & Posterior Pituitary

ADH is synthetised in the hypothalamus & is transported to the posterior pituitary.

ADH is a nonapeptide produced in the supraoptic and paraventricular nuclei and other areas of the hypothalamus. Its major role is in the regulation of water balance by its effect on the kidneys. ADH is also known as vasopressin because of the vasopressor response to pharmacological doses. Humans and most animals have arginine-vasopressin but pigs have the arginine replaced by a lysine.

ADH is produced from a much larger precursor protein (prepropressophysin). The gene for this precursor is located on human chromosome 20 and is very closely related to the oxytocin gene. These genes probably arose from an ancestral gene as a result of gene duplication about 350 million years ago. The ADH precursor protein contains sequences for three separate peptides into which it is split during transport down the nerve axon to the posterior pituitary. These are ADH, neurophysin & a glycopeptide. The physiological role of these later two peptides is unclear but neurophysin may have a role as a carrier or binding protein within the granules.

The secretory granules containing the ADH and neurophysin move down the axons (axonal transport) to the nerve terminals in the posterior pituitary from where they are secreted into the systemic circulation by a process of exocytosis (involving calcium).

Intravascular ADH has a half-life of only about 15 minutes being rapidly metabolised in the liver and kidney to inactive products.

5.6.2: Renal Actions of ADH

ADH acts on receptors in the basolateral membrane of cells in the cortical and medullary collecting tubules and not on the apical (or luminal) membrane. These membranes have different properties. The apical membrane of these cells is impermeable to water in the absence of ADH but the basolateral membrane is always permeable to water.

ADH initiates its physiological actions by combining with a specific receptor. These are two major types of vasopressin receptors: V1 & V2. The V1 receptors are located on blood vessels and are responsible for the vasopressor action.

The V2 receptors are in the basolateral membrane of the collecting tubule cells in the kidney. Various agonists and antagonists at these receptors have been developed. Desamino-d-arginine vasopressin (dDAVP) is a synthetic V2-agonist which is used clinically in treatment of diabetes insipidus.

The action at the V2 receptor activates adenyl cyclase and cyclic AMP (second messenger) is formed. This initiates a series of events which causes specific vesicles in the cytoplasm to move to and fuse with the apical membrane. The vesicles contain the water channels (aquaporin 2) which are now inserted in the apical (ie luminal) membrane rendering it permeable to water. Water moves into the cell through these channels in response to the osmotic gradient. It passes into the circulation across the basolateral membrane. The basolateral membrane is always freely permeable to water but the apical membrane is permeable only when the water channels are inserted. When intracellular cyclic AMP levels fall, the water channels are removed from the membrane and reform as vesicles.

The cycle of insertion of water channels into then removal from the luminal membrane is referred to as vesicular trafficking and is the final mediator of the ADH-dependent water permeability of the collecting duct cells.

The water channels are membrane proteins called aquaporins. Aquaporin-2 is the protein which is the vasopressin responsive water channel in the collecting duct. It is inserted into the apical membrane in reponse to cyclic AMP. The protein forms a tetrameric complex that spans the membrane and forms a channel which allows rapid water movement in response to an osmolar gradient.

Aquaporins 3 & 4 are the water channels located in the basolateral membrane. Their water permeability is not altered by ADH action and their presence means the basolateral membrane has a continuous water permeability.

Other interesting recent findings in this area are:

- Mercurial diuretics bind to a specific site on aquaporin-2 and block water reabsorption. This is the mechanism of their diuretic action
- The autosomal dominant form of nephrogenic diabetes insipidus is due to mutations in the aquaporin-2 gene

- The X-linked form of nephrogenic diabetes is due to mutations in the gene for the V2 vasopressin receptor. (This receptor gene is on the X chromosome)
- Lithium causes marked down-regulation of aquaporin-2 expression and causes a form of acquired nephrogenic diabetes insipidus

Overall Effects in the Kidney

- In the absence of ADH, the apical membranes of the cells in the cortical and medullary collecting tubules have very low water permeability. Large volumes of hypotonic urine are produced. Up to 12% of the filtered load of 180l/day is excreted (urine volume up to 23 liters/day!)
- In the presence of ADH, the cells are much more permeable to water. At maximal ADH levels, less then 1% of the filtered water is excreted (urine volume 500mls/day)
- Feedback loop: Reabsorption of water reduces plasma [Na^+] and this is detected by the osmoreceptors in the hypothalamus. This allows sensitive feedback control of ADH secretion. (Aquaporin 4 is found in the cells of the thirst centre in the hypothalamus and is probably involved in the mechanism which monitors plasma tonicity)

5.8: Coupling of the Osmoreceptor and the Kidneys

The role of the hypothalamic osmoreceptor in the control of water balance has been experimentally determined and the quantitative aspects of its function will be discussed here.

A graph of plasma ADH levels versus plasma osmolality is depicted below. The points to note are:

- Below an osmolality of about 280 mOsm/l, ADH levels are very low
- The curve starts to rise very sharply and linearly at osmolalities above 280 mOsm/l

The value of 280 mOsm/l is the threshold value (or set-point) of the osmoreceptor. The slope of the line above 280 mOsm/l represents the sensitivity of the receptor. This rising line can be described by the equation:

$$[ADH] = 0.38 \times (POsm - 280)$$

where: [ADH] is the plasma ADH concentration and POsm is the plasma osmolality.

The sensitivity (the slope of the line) is 0.38 pg of ADH/ml per mOsm/kg. [ADH] will increase by 0.38 pg/ml for every 1 mOsm/kg increase in plasma osmolality.

The extreme sensitivity of the osmoreceptor can be better appreciated if this is stated another way: A 1% increase (2.9 mosm/kg) in plasma osmolality will increase [ADH] by about 1 pg/ml. This increase is enough to have significant effects on urinary osmolality.

The next step is to consider the relationship between the [ADH] and the urine osmolality. This is displayed in the figure below. As the [ADH] increases the antidiuretic effect increases and the urine osmolality increases up to the limit set by the maximal concentrating ability of the kidney. For young adults, this maximal urine osmolality is somewhere between 1200 to 1400 mOsm/kg. The line can be described by the equation:

$$UOsm = 250([ADH] = 0.25)$$

where: [ADH] is the plasma ADH concentration and UOsm is the urine osmolality.

The slope of the line (250) is the sensitivity of the renal mechanism which responds to ADH. The threshold [ADH] is 0.25 pg/ml. To state this another way: An increase in [ADH] of just 1 pg/ml will cause urine osmolality to rise by 250 mOsm/kg. The renal response to ADH is very sensitive.

The overall sensitivity of this system for controlling plasma osmolality and water balance is referred to as the gain of the system. The gain is high because there is a sensitive mechanism for responding to changes in plasma osmolality which is coupled to a sensitive mechanism for changing urine osmolality in response to changes in [ADH].

The sensitivity of the osmoreceptor (0.38) is such that a rise in plasma osmolality of 2.63 mOsm/kg (ie $\frac{1}{0.38}$) will result in a rise of 1 pg/ml in [ADH]. The sensitivity of the renal response is 250 so the overall gain of the system is 95 (ie $\frac{250}{2.63}$). This means that an increase in plasma osmolality of 1 mOsm/kg will result in a rise in urine osmolality of 95 mOsm/kg!

The two limits imposed on the system must be recognised:

- the threshold of the osmoreceptor (280 mOsm/kg)
- the maximal urine concentration of the kidney (1200 to 1400 mOsm/kg in a young adult)

The sensitivity of the system is so high that it exceeds our ability to accurately measure the osmolality.

The [ADH] at the threshold of the osmoreceptor is 0.5 pg/ml. The [ADH] at the maximal urine concentration is 5 pg/ml. The plasma osmolality in healthy adults averages 287 mOsm/kg and this is associated with a [ADH] of 2 to 2.5 pg/ml. The significance of this is that it is at about the midpoint of the renal response line: sensitivity to changes in plasma osmolality is thus high in both directions.

Maximal anti-diuresis occurs at a plasma osmolality of 294 mOsm/kg. This is about the average osmolality at which the thirst mechanism is activated. This illustrates the interaction btween the ADH and the thirst mechanisms for control of water balance. The threshold of thirst for osmotic stimuli has a higher set-point then that for ADH release: thirst is considered by some to act as a back-up mechanism if changes in ADH are not sufficient of themselves to keep plasma osmolality constant.

At the threshold of the osmoreceptor (280 mOsm/kg), the [ADH] is less than 0.5 pg/ml and urine osmolality is at its minimal value. The formula predicts minimal urine osmolality of about 60 mOsm/kg (ie $250 \times (0.5 - 0.25)$)if basal [ADH] is 0.5 pg/ml. The minimum urine osmolality that is measured in young adults is in the range 40 to 100 mOsm/kg. To excrete a daily solute load of 600 mOsm at a minimum urine osmolality of 60 mOsm/kg requires a urine volume of 10 liters (over 400 mls/hr). The significance of this is that the urine production can increase to such high levels when [ADH] is low, that hypotonic hyponatraemia cannot persist other then briefly if the kidney's ability to excrete dilute urine is normal.

The response of the osmoreceptor can be affected by several other factors:

- Intravascular volume
- The rate of change of the osmolality
- The type of blood solute present

The response of the osmoreceptor is partially rate-dependent: a rapid increase in plasma osmolality will result in a much higher [ADH] initially then if the plasma osmolality has risen slowly. This effect is noticeable if the plasma osmolality increases at a rate of 2% or more per hour.

Some blood solutes are less effective than others in stimulating the osmoreceptor. Sodium and its associated anions normally account for about 92% of plasma tonicity; so under normal conditions the osmoreceptor responds to changes in sodium concentration. Glucose & urea contribute to plasma osmolality but they cross cell membranes easily and are ineffective solutes which do not contribute to plasma tonicity (see **Section 1.2.3**). An increase in urea concentration can have marked effects on plasma osmolality but minimal effects on blood tonicity and thus does not affect [ADH]. The osmoreceptor senses blood tonicity and not blood osmolality. In the presence of insulin, glucose can enter the osmoreceptor cells and is an ineffective osmole. In cases of hyperglycaemia due to insulin deficiency, glucose cannot enter cells so it now is effective in altering plasma tonicity and can cause appropriate release of ADH.

5.9: Interaction between Volume and Osmolality in Control

Changes in blood volume or blood pressure have major effects on osmoreceptor function. The threshold and the sensitivity of the osmoreceptor can both be altered. Overall, the change that occurs is such that it would tend to rapidly correct the disturbance (provided of course that the kidney was able to respond to [ADH] changes normally). Experiments in humans show that blood volume changes do not have much effect until the change in blood volume is of the order of 7 to 10%. Changes in blood pressure have similar effects and this presumably is sensed via the carotid baroreceptor mechanism.

The change with hypovolaemia increases the ADH response to a given level of osmolality. This causes renal water retention to assist with correcting the hypovolaemia but this occurs at the expense of maintaining the normal plasma osmolality. Hypovolaemia impairs water excretion & tends to cause hypotonic hyponatraemia.

5.10: Other Mechanisms in Water Balance

There are other renal mechanisms which can have major effects on water excretion and which act independently of the thirst-ADH effector system discussed above. These are additional effector mechanisms which are important and which all act to alter renal water or sodium excretion.

The major additional mechanisms which act at the local renal level are:

- Glomerulotubular Balance
- Autoregulation
- Intrinsic Pressure-Volume Control System
- Natriuretic peptides

Glomerulotubular Balance

Glomerulotubular balance is a strictly local renal mechanism. It refers to the finding that the proximal tubule tends to reabsorb a constant proportion of the glomerular filtrate rather than a constant amount. The effect of this is to minimise the effect of changes in GFR on sodium and water excretion.

How does this mechanism work? This is not completely understood but there are probably several factors involved. Changes in oncotic pressure are undoubtably important and the mechanism can be understood from a consideration of this factor alone. Changes in hydrostatic pressure & in delivery of certain solutes to the proximal tubule are probably also involved.

When GFR increases, the protein concentration (& oncotic pressure) in the efferent arteriole is immediately increased resulting in increased oncotic pressure in the peritubular capillaries. This results in an increased gradient favouring reabsorption and counteracts (balances) the effect of an increased GFR on volume of fluid leaving the proximal tubule. This is a self-regulating mechanism acting locally. It has effects on water excretion if the oncotic pressure of plasma is lowered.

Autoregulation of Renal Blood Flow

Autoregulation of renal blood flow is another local renal mechanism which has effects on water excretion. If the renal perfusion pressure increases, the afferent arterioles vasoconstrict so that renal plasma flow (RPF) and GFR are maintained constant. The mechanism of this pressure autoregulation is not understood but may be due to a local myogenic response (ie. the vascular smooth muscle of the afferent arteriole may respond to the increased stretch by contracting and increasing afferent arteriolar resistance). Most likely however, other renal mechanisms such as tubulo-glomerular feedback are important.

RPF and GFR are autoregulated and kept fairly constant and this greatly minimises the effect of changes in BP on urine output. However urine flow is not autoregulated! An increase in blood pressure will cause an increase in urine flow even though GFR is minimally affected. How can this be so? The increase in GFR is small but may still result in a significant increase in urine flow even though most of the effect of the increased GFR is buffered by the glomerulotubular balance mechanism. This is a local renal mechanism which is of major importance in maintenance of a constant intravascular volume. The altered blood pressure will also have effects on ADH secretion via carotid baroreceptor input and this will affect water excretion in the same direction as the local renal mechanism.

Intrinsic Pressure-Volume Control System

The pressure-volume control system mentioned above is the intrinsic control system for maintaining a constant blood volume.

Pressure diuresis
Pressure natriuresis

Natriuretic Peptides

to be completed

[add details to this section]

CHAPTER OVERVIEW

6: Control of Compartment Volumes

6.1: Osmotic Forces

Osmosis refers to water flow across a membrane into a region where there is a higher concentration of a solute to which the membrane is impermeable. Water moves because of diffusion down a concentration gradient. All fluid compartments in the body are isotonic as water movement across cell membranes occurs rapidly and easily. The resulting distribution of water that occurs between the compartments is essentially the result of this water movement across membranes.

What determines the distribution of the total body water between the ICF & the ECF?

Assume for the moment that cells contain a constant amount of solute which gives the ICF a certain tonicity. Water can cross cell membranes readily so:

Intracellular tonicity must always equal ECF tonicity.

If cell solute is constant than the ECF tonicity (which may vary) determines how much water will enter the cell. Water enters until the osmolar gradient is abolished. The extracellular tonicity determines the relative distribution of the total body water between the ICF and the ECF. If ECF tonicity increased, then water would move out of the cell and extracellular volume would increase at the expense of intracellular volume. This is the basis of using a hypertonic infusion such as 20% mannitol to decrease intracellular volume: this effect will occur in all cells but the target organ is usually the brain. If ECF tonicity decreased, the reverse situation applies.

What determines ECF tonicity? Na+ and obligatorily associated anions account for about 92% of ECF tonicity. Na^+ is an effective osmole across the cell membrane because of its low membrane permeability and the sodium pump which together effectively exclude ECF Na+ from the ICF. The relative volumes (ie distribution) of water between the ICF and the ECF can be considered as being determined by the ECF [Na^+]!

That is: If intracellular solute content is constant then:

The distribution of the TBW between the ECF and the ICF is determined by the ECF [Na^+].

For example, if ECF [Na^+] rises (at constant total body water), then ECF volume increases (and ICF volume decreases by the same amount).

The assumption that intracellular content is constant is not always correct (discussed in Section 6.2) but these special circumstances do not greatly detract from the general conclusion here.

What determines the distribution of the ECF between the IVF & the ISF?

The other major fluid division is between intravascular fluid and interstitial fluid. The capillary membrane is the relevant semi-permeable membrane to consider here. Water and electrolytes can all readily cross this membrane. All the electrolytes and other small molecular species are ineffective at exerting an osmotic force across this membrane.

Plasma contains a small amount of large molecular weight particles (colloids, mostly proteins) which contribute only about half a percent of the total osmolality of plasma. These proteins have only a very limited permeability across the capillary membrane. As the proteins are the only compounds capable of exerting an osmotic force across the capillary membrane, they account for all the osmotic force exerted across this membrane. The fact that the protein concentration of the ISF is lower means that there is an osmotic gradient across the capillary membrane. This gradient is usually referred to as an oncotic pressure gradient. The term tonicity is rarely used in this context. because of possible confusion because tonicity is usually discussed in relation to the cell membrane. This oncotic gradient along with the hydrostatic pressure gradient are the major determinants of the relative distribution of the ECF between plasma and ISF. This concept is referred to as Starling's hypothesis.

Summary: Some Rules of Water Control in the Body

1. Water crosses (most) cell membranes easily

2. Intracellular osmolality must always equal extracellular osmolality

3. Extracellular osmolality is effectively determined by the ECF [Na^+]

4. ECF [Na^+] determines ICF volume

5. Osmoreceptor control of osmolality is sensitive and powerful so ECF [Na^+] is held constant

6. Total body solute is relatively constant

6.2: Regulation of Cell Volume

Most cell membranes are freely permeable to water

As most cell membranes are freely permeable to water and do not possess water pumps in their membranes, cells will shrink or swell in response to changes in ECF tonicity. This is generally undesirable for most cells which need a constant cell volume to maintain optimum function.

Among the few exceptions to this rule are the epithelium of the urinary bladder and certain special cells and segments in the renal tubule.

How do cells respond to these extracellular osmotic stresses and maintain a constant volume?

Cells contain a significant concentration of large molecular weight anionic colloids (mostly proteins and organic phosphates) which cannot cross the cell membrane. In contrast, interstitial fluid generally has a low protein concentration. The high intracellular concentration of non-diffusable anions leads to a Gibbs-Donnan equilibrium across the cell membrane. At equilibrium (if it occurred), electroneutrality would be preserved on both sides of the membrane but there would be more particles (higher osmolality) intracellularly. Water would enter the cell down its concentration gradient and the cell would tend to swell. This would upset the Gibbs-Donnan equilibrium and more solute particles would enter the cell which would swell even more ... ... and so on. This is an unstable situation which, if unopposed, would lead to cell rupture.

How can this be as we know that cell volume tends to be very stable? The above argument is valid and applicable to all cells. What is the mechanism which prevents cell swelling and rupture? The answer is the sodium pump (Na^+-K^+ ATPase) in the cell membrane. The pump together with the membrane's low permeability to sodium, effectively prevents sodium from entering the cell. The sodium becomes an extracellular cation to which the membrane is effectively impermeable. This sets up another Gibbs-Donnan equilibrium now with Na^+ as the impermeable charged species.

Overall, the equilibrium situation is that the Gibbs-Donnan effect due to the impermeant extracellular sodium balances the Gibbs-Donnan effect due to the impermeant intracellular colloids. This double-Donnan effect stabilises cell volume.

If the sodium pump was blocked (eg by drugs), sodium would enter the cell and water would follow until the cell ruptured. The sodium pump is important in stabilising cell volume in addition to its critical role in the generation of the resting membrane potential.

In summary so far

- Intracellular colloid (mostly proteins and organic phosphates) cannot cross the cell membrane. These anions affect the distribution of the diffusable ions according to the Gibbs-Donnan effect
- The sodium pump renders the membrane effectively impermeable to sodium: this sets up another Gibbs-Donnan equilibrium which has effects opposite to the first
- The balance between these two effects allows the cell to maintain a normal cell volume

What happens to cell volume when cells are stressed by a change in ECF tonicity?

Water crosses membranes freely, so this change in tonicity will have rapid (several minutes) effects on cell volume. A hypertonic ECF will cause cells to shrink; a hypotonic ECF will cause cells to swell. This is undesirable for normal cell function and this is especially disadvantageous in the brain.

On acute exposure to a hypotonic ECF, cells do swell within a couple of minutes but then their volume starts to decrease towards normal. This decrease is termed volume regulatory decrease and is due to loss of intracellular solute particularly potassium.

In hypertonic ECF, cells decrease in size but are able to partially recover: this is termed volume regulatory increase and acutely is due to a net leak of solute (mostly Na^+ and Cl^-) into the cell.

If the ECF tonicity is only slowly changed, then the response of the cell is different. The cells are able to adapt as the tonicity is changed. They are able to minimise any change in cell volume over a wide range of osmolality. This happens because the cell is able to lose or gain solute at a rate which almost matches the effect of the change in tonicity.

If a cell which has partially recovered towards its normal cell volume is suddenly returned to a situation of normal ECF tonicity, then the reverse effect occurs eg a swollen cell which has lost solute and decreased its cell volume will shrink markedly if suddenly returned to normal ECF tonicity. This is the predictable outcome based on the lowered intracellular tonicity responsible for the return of volume towards normal.

An example of this is the difference in symptomatology of acute hyponatraemia versus chronic hyponatraemia. For the same absolute plasma [Na+], chronic hyponatraemia is much better tolerated than acute hyponatraemia. The brain cells in chronic hyponatraemia have reduced their cell volume and significantly restored their normal functioning. The converse holds for rapid correction of the hyponatraemia. Rapid normalisation of ECF tonicity in chronic hyponatraemia can result in marked symptoms due to rapid decrease in cell size; but rapid correction of acute hyponatraemia may be much better tolerated.

Idiogenic Osmoles

For many types of cells an additional very important mechanism is operative. Consider the brain which has been subjected to a hypertonic ECF. The brain cells may gain solute (principally Na^+ and Cl^-) from the extracellular environment and return their volume towards normal. However, the brain cells are capable of increasing their tonicity by gaining solute using another mechanism. They can produce more particles from cellular metabolism. These substances are known as idiogenic osmoles (or osmolytes) and include taurine, glycine, glutamine, sorbitol and inositol. An increase in these idiogenic osmoles have been detected in brain cells as early as 4 hours after an acute hypertonic challenge.

The production of extra osmoles within the cell is very important. The problem with taking in inorganic ions like Na^+ and Cl^- from the ECF is that higher than normal concentrations of these ions have adverse effects on intracellular enzyme systems. Coping with intracellular dehydration is a problem common to many animals. The evolutionary response has been to allow cells to generate extra osmoles inside the cell by producing certain compounds which do not disrupt enzyme function. These idiogenic osmoles have also been termed compatible osmoles because of their relatively benign effect on intracellular proteins.

Points to Note

- The kidney is the major regulator of ECF tonicity (in response to sensitive osmoreceptor monitoring and ADH activity)
- Normally ECF tonicity is relatively constant and this maintains the volume of all cells in the body (and thus determines total intracellular volume and the distribution of total body water between ICF and ECF).
- All cells have their own local mechanisms which attempt to maintain a constant cell volume. (The sodium pump is critically important in rendering the cell effectively impermeable to sodium and counteracting the Gibbs-Donnan effect of the intracellular colloids. This maintains a normal cell volume under isotonic conditions).
- In situations of osmotic stress, cells attempt to return their cell volume to normal by either gaining or losing intracellular solute.
- Extra intracellular solute may come from ECF solute (more disruptive to cell function) or from metabolic generation of extra idiogenic solute (more compatible with cell function)
- These volume regulatory processes operate at the level of the individual cell and protect the cells from the volume changes that would occur due to changes in ECF [Na^+] (tonicity).

These cellular events have great significance if rapid correction of a chronic osmolar disturbance is attempted. Rapid normalisation of chronic hyponatraemia can cause severe neurological symptoms.

Remember also that plasma [Na^+] is an *index of water balance* rather than of sodium balance and is regulated by the processes which control water balance (ie the thirst-ADH mechanism).

Brief Overview

- Kidneys regulate ECF [Na^+]
- ECF [Na^+] controls the distribution of water between ECF and ICF at any instant
- Cells can also regulate their own cellular volume by changing intracellular solute content to minimise the adverse functional effects of changes in ECF tonicity

6.3: Blood Volume Control

To be completed

CHAPTER OVERVIEW

7: Intravenous Fluids

7.1: Classification

To be completed

7.2: Crystalloids

Why use crystalloids?

The advantages of crystalloid solutions are:

- inexpensive
- easy to store with long shelf life
- readily available
- very low incidence of adverse reactions
- a variety of formulations are available
- effective for use as replacement fluids or maintenance fluids
- no special compatibility testing is required.
- no religious objections to their use

In essence they are cheap and effective and don't cause adverse reactions.

Crystalloids solutions are classified into three groups based on their predominant use. The contents of the various solutions are listed in the table below.

Replacement Solutions

These solutions are used to replace ECF. They are all isotonic. The key factor is that these solutions have a [Na^+] similar to that of the extracellular fluid which effectively limits their fluid distribution to the ECF. The fluid distributes between the ISF and the plasma in proportion to their volumes. Intracellular fluid volume does not change. If used to replace blood loss, 3 to 4 times the volume lost must be administered as only 1/3 to 1/4 remains intravascularly. In healthy adults with a normal initial haemoglobin level, up to 20% loss of blood volume (loss of approx 1,000 mls) can be safely replaced with a 3,000-4,000 ml infusion of replacement solution without any adverse effects.

Hartmann's solution contains lactate as a bicarbonate precursor. The metabolism of lactate in the liver results in production of an equivalent amount of bicarbonate. Similarly, Plasmalyte 148 solution contains acetate and gluconate as bicarbonate precursors. These anions (eg lactate) are the conjugate base to the corresponding acid (eg lactic acid) and do not contribute to development of an acidosis as they are administered with Na^+ rather than H^+ as the cation.

Maintenance Solutions

These solutions are used to provide maintenance fluids. They are isosmotic as administered and do not cause haemolysis. Following administration, the glucose is rapidly taken up by cells so the net effect is of administering pure water. Dextrose 5% contains no Na^+ so it is distributed throughout the total body water with each compartment receiving fluid in proportion to its contribution to the TBW. (See Section 8.1).

Some maintenance solutions also have Na^+ so they can be prescribed to provide the daily maintenance requirements for water and Na^+. Supplemental K^+ can be added as required. The normal daily Na^+ intake of 1.5 to 2 mmol/kg can be given in this way by appropriate fluid selection. The Na^+ content does alter the fluid distribution but this is predictable.

Hartmann's solution contains Ca^{2+} and this can cause problems if administered with stored blood. Citrate is the anticoagulant used in stored blood and it works by complexing with and removing Ca^{2+} from solution. It is possible for the Ca^{2+} in Hartmann's to cause clotting of blood in the infusion tubing particularly if the blood is given slowly or the tubing contains reservoir areas (eg as in pump sets). For this reason, it has become standard practice to administer normal saline before and after a blood transfusion to prevent blood and Ca^{2+} mixing in the infusion tubing. Plasmalyte 148 solution contains Mg^{2+} instead of Ca^{2+} and can be administered with stored blood without causing this problem.

Special Solutions

Some crystalloid solutions used for special purposes are grouped together here, for example:

- Hypertonic (3%) saline

- Half normal saline
- 8.4% Bicarbonate solution
- Mannitol 20%

7.3: Colloids

Colloids are large molecular weight (nominally MW > 30,000) substances. In normal plasma, the plasma proteins are the major colloids present. Colloids are important in capillary fluid dynamics because they are the only constituents which are effective at exerting an osmotic force across the wall of the capillaries. Albumin solutions are available for use as colloids. In addition, various other solutions containing artificial colloids are available. The general problems with colloid solutions are:

- much higher cost than crystalloid solutions
- small but significance incidence of adverse reactions (esp anaphylactoid reactions)

7.3.1: Molecular Weight

Two molecular weights are quoted for colloid solutions (see Huskisson 1992 for definitions):

- Mw : Weight average molecular weight
- Mn : Number average molecular weight

The Mw determines the viscosity and Mn indicates the oncotic pressure. Albumin is said to be monodisperse because all molecules have the same molecular weight (so $Mw = Mn$). Articial colloids are all polydisperse with molecules of a range of molecular weights.

7.3.2: The Ideal Colloid Solution

The properties of an ideal colloid solution for use as a plasma volume expander are outlined in the table. An oncotic pressure similar to plasma will permit replacement of plasma volume without distribution to other fluid compartments and this is the key element that makes a solution a colloid solution.

Table 7.3: The Properties of an Ideal Colloid

General

- Distributed to intravascular compartment only
- Readily available
- Long shelf life
- Inexpensive
- No special storage or infusion requirements
- No special limitations on volume that can be infused
- No interference with blood grouping or cross-matching
- Acceptable to all patients & no religious objections to its use

Physical Properties

- Iso-oncotic with plasma
- Isotonic
- Low viscosity
- Contamination easy to detect

Pharmacokinetic Properties

- Half-life should be 6 to 12 hours
- Should be metabolised or excreted & not stored in the body

Non-Toxic & No Adverse Affect on Body Systems

- No interference with organ function even with repeated administration
- Non-pyrogenic, non-allergenic & non-antigenic
- No interference with haemostasis or coagulation
- Not cause agglutination or damage blood cells
- No affect on immune function including resistance to infection
- No affect on haemopoiesis
- Not cause acid-base disorders
- Not cause or promote infection (bacterial, viral or protozoal)

7.3.3: Dextrans

Dextrans are highly branched poysaccharide molecules which are available for use as an artificial colloid. They are produced by synthesis using the bacterial enzyme dextran sucrase from the bacterium Leuconostoc mesenteroides (B512 strain) which is growing in a sucrose medium.

The formulations currently available are:

> Dextran 40 (Mw 40,000 & Mn 25,000) [Rheomacrodex]
>
> Dextran 70 (Mw 70,000 & Mn 39,000) [Macrodex].

The dextrans cause more severe anaphylactic reactions than the gelatins or the starches. The reactions are due to dextran reactive antibodies which trigger the release of vasoactive mediators. Incidence of reactions can be reduced by pretreatment with a hapten (Dextran 1).

Dextran 70 has a duration of action of 6 to 8 hours. Interference with crossmatching occurs so the laboratory should be notified that dextrans have been used. Dextran interferes with haemostasis; it induces an acquired von Willebrand's state. Consequently, there is a maximal dosage recommendation of 20 mls/kg (about 1,500 mls in an adult).

Dextran40 is used to improve microcirculatory flow in association with certain procedures (eg microsurgical reimplantations).

7.3.4: Gelatins

Gelatin is the name given to the proteins formed when the connective tissues of animals are boiled. They have the property of dissolving in hot water and forming a jelly when cooled. Gelatin is thus a large molecular weight protein formed from hydrolysis of collagen.

Gelatin solutions were first used as colloids in man in 1915. The early solutions had a high molecular weight (about 100,000). This had the advantage of a significant oncotic effect but the disadvantages of a high viscosity and a tendency to gel and solidify if stored at low temperatures. Reducing the molecular weight reduced the tendency to gel but smaller molecular weight molecules could not exert a significant oncotic effect. The need was for a modified gelation product that had a moderate molecular weight (for oncotic pressure) but a low gel melting point. (It is difficult to infuse a jelly).

Several modified gelatin products are now available; they have been collectively called the New-generations Gelatins. There are 3 types of gelatin solutions currently in use in the world:

- Succinylated or modified fluid gelatins (eg Gelofusine, Plasmagel, Plasmion)
- Urea-crosslinked gelatins (eg Polygeline)
- Oxypolygelatins (eg Gelifundol)

Polygeline (Haemaccel Hoechst) is available in Australia. The gelatin is produced by the action of alkali and then boiling water (thermal degradation) on collagen from cattle bones. The resultant polypeptides (MW 12,000 - 15,000) are urea-crosslinked using hexamethyl di-isocyanate. The branching of the molecules lowers the gel melting point. The MW ranges from 5,000 to 50,000 with a weight-average MW of 35,000 and a number-average MW of 24,500.

Properties

Polygeline is supplied as a 3.5% solution of degraded gelatin polypeptides cross-linked via urea bridges with electrolytes (Na^+ 145, K^+ 5.1, Ca^{2+} 6.25 & Cl^- 145 mmol/l). It is sterile, pyrogen free, contains no preservatives and has a recommended shelf-life of 3 years when stored at temperatures less than 30°C.

Handling by the Body

It is rapidly excreted by the kidney. Following infusion, its peak plasma concentration falls by half in 2.5 hours. Distribution (as a percent of total dose administered) by 24 hours is 71% in the urine, 16% extravascular and 13% in plasma The amount metabolised is low: perhaps 3%.

Indications

The major use of Polygeline is the replacement of intravascular volume eg correcting hypovolaemia due to acute blood loss. It is also used in priming heart-lung machines.

Advantages

- Lower infusion volume required as compared to crystalloids
- Cheaper and more readily available then plasma protein solutions
- No infection risk from the product if stored and administered correctly
- Only limit to the volume infused is the need to maintain a certain minimum [Hb] (In comparison, dextrans have a 20ml/kg limit).
- Readily excreted by renal mechanisms
- Favourable storage characteristics: long shelf life, no refrigeration
- No interference with blood cross-matching
- Compatible with other IV fluids except Ca^{2+} can cause problems with citrated blood products.

Disadvantages

- Higher cost then crystalloids
- Anaphylactoid reactions can occur
- No coagulation factors and its use contributes to dilutional coagulopathy

Starches

These polydisperse colloid solutions are produced from amylopectin which has been stabilised by hydroxyethylation to prevent rapid hydrolysis by amylase. Hydroxyethylstarch is removed from the circulation by renal excretion and by redistribution. Anaphylactoid reactions occur in about 0.09% of cases. Some patients experience severe pruritis. A particular concern is the possibility that starch preparations can affect the coagulation process. This issue has not been resolved but it seems prudent to avoid its use in patients with a coagulopathy. The maximum recommended dose is 20 mls/kg so its use in major resuscitation is limited

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CHAPTER OVERVIEW

8: Applied Physiology of Transfused Fluids

8.1: Infusion of Isomolar Fluids

Consider the Distribution & Excretion of 1,000 mls of various fluids.

As an exercise in applied fluid physiology is to compare the distribution and excretion of a rapid intravenous infusion of 1000 mls of various fluids. This serves to emphasise some of the factors involved in the selection of an appropriate fluid. In this exercise we will tend to ignore cardiovascular changes such as increased venous capacitance.

Certain simplifing assumptions are made in clinical practice about the sizes of the various fluid compartments to facilitate the mental arithmetic without loss of any clinically relevant precision. The water of dense connective tissue & bone is significant in volume (15% of total body water) but as a kinetically slow compartment. It is not important in consideration of short term fluid distribution. Transcellular fluids are small in volume and usually slow so they too are excluded from this clinical analysis.

This leaves three big compartments:

- Intracellular fluid (55% of TBW, 23 liters)
- Interstitial fluid (20% of TBW, 8.4 liters)
- Intravascular fluid (Plasma 7.5% of TBW, 3.2 liters and Red cell volume 1.8 liters).

The IVF is the blood volume with 5 liters in total. The red cell volume is part of the ICF but also is part of the blood volume. The ratio of ICF to ECF is 23:11.6 (about 2:1). The ratio of ISF to plasma volume is 8.4:3.2 and this will be treated as about 3:1. This discussion only considers those parts of the total body water that are rapidly equilibrating. These are the only components that need to be considered in acute fluid changes.

Assumptions used for this Simple Analysis

- TBW is one-third ECF & two-thirds ICF
- ECF is one-quarter plasma & three-quarters ISF
- The threshold of the volume receptors is 7-10% change in blood volume
- The osmoreceptors are sensitive to a 1-2% change in osmolality.
- Plasma osmolality is normal prior to the transfusion (ie 287-290 mOsm/kg)

Now, consider the rapid IV administration of 1,000 mls of the following fluids: Dextrose 5%, normal saline and plasma protein solution. The type of questions to be considered are:

How are these different fluids distributed in the body?

How are tonicity and intravascular volume affected?

What are the mechanisms the body uses to excrete these fluids?

Which is excreted the most rapidly?

Dextrose 5%

Dextrose 5% is a Maintenance Fluid. (Dextrose is d-glucose). It is isosmotic as administered and does not cause haemolysis. The glucose is rapidly taken up by cells. The net effect is of administering pure water, so it is distributed throughout the total body water. Each compartment receives fluid in proportion to its contribution to the TBW (ie 2/3rd to ICF and 1/3rd to ECF; the ECF fluid is distributed one quarte to plasma & three quarters to ISF).

The distribution of 1,000 mls of dextrose 5% is:

- ICF 670mls
- ECF 330mls (with ISF 250mls and plasma 80mls).

(The figures are rounded slightly)

Intravascular volume increases from 5000 to 5080 mls. This volume increase of less then 2% which will not be sensed by the volume receptors (as it is below the 7-10% threshold).

The osmolality of plasma (3,200 mls) will decrease by: [$287 - \frac{287 \times 3.20}{3.28}$] which is about 7 mOsmoles/l or a 2.5% decrease. This is enough to be detected by the osmoreceptors. ADH release will be decreased and renal water excretion will rise. A delay will occur because the changes have to be detected centrally and then ADH levels need 3 half-lives to fall to a new steady state.

Normal Saline

Normal saline is an ECF Replacement Fluid. Its [Na^+] is similar to that of the extracellular fluid and this effectively limits its distribution to the ECF (distributing between the ISF & the plasma in proportion to their volume ie 3:1).

The ISF will increase in volume by 750 mls. The plasma volume will increase by 250 mls. This is why blood loss of 1,000 mls requires about 3 to 4 times the volume of IV replacement fluid to restore normal intravascular volume.

Plasma osmolality and tonicity will be unchanged because normal saline is isosmotic. The osmoreceptors do not contribute anything to the excretion of normal saline. Blood volume increases to 5250 mls; an increase of 5%. This is below the sensitivity of the volume receptors. It seems that the body has no clear way of excreting this excess fluid as neither osmoreceptors nor volume receptors are stimulated! However, experiments have shown that replacement fluids are excreted the most rapidly of all these groups!

How does this happen? An additional mechanism is relevant here. Normal saline contains no protein so the oncotic pressure in the blood is slightly lowered following the saline infusion. This has 2 effects:

- Movement of fluid into the ISF is favoured (Starling's Hypothesis)
- Glomerulo-tubular imbalance occurs: the lowered oncotic pressure immediately leads to an increase in GFR and a smaller reabsorption of water in the proximal tubule. Urine flow increases. This is a strictly local effect without any hormonal intermediary. The urine flow increases immediately. Fluid then moves back into the intravascular compartment and the urine flow continues until all the transfused fluid is excreted.

Plasma Protein Solution

Plasma protein solution is a colloid and is distributed only to the intravascular fluid. The tonicity is unaltered. The blood volume increases from 5,000 mls to 6,000 mls; an increase of 20%. This is above the 7 to 10% threshold for the volume receptors. The result is a fall in ADH levels and the excretion of the excess water commences.

This water loss tends to increase the plasma oncotic pressure and water moves from the ISF to the IVF. Vascular reflexes are important also in causing venous pooling and a decrease in the effective circulating volume. These mechanisms tend to slow the excretion of the water load. The albumin is partly slowly redistributed to the ISF and metabolised. These changes are slow so the effect of plasma protein infusion on blood volume is both more pronounced and more prolonged.

The pressure-volume control mechanisms important in long term regulation of blood volume are slow in onset but become relevant here as the blood volume change is more significant and more prolonged and occurs without change in osmolality (or initially in plasma oncotic pressure either).

Overview

Dextrose 5% is essentially treated by the body as pure water and a significant percent moves intracellularly. It is a useful fluid to replenish intracellular fluid but does so at the expense of tonicity. It is inappropriate for intravascular volume replacement. It is excreted because ADH levels decline in response to the drop in plasma osmolality.

Normal saline is a replacement fluid (meaning ECF replacement) because it adds only to the ECF volume. Only about a third remains intravascularly. To replace intravascular volume will require transfusion with about 3 times the volume of blood lost. It is cheap and readily available. It is excreted because the small drop in plasma oncotic pressure causes glomerulotubular imbalance. ADH is not affected.

Plasma protein solutions (eg 5% human albumin) are excellent for replacing intravascular volume. ISF and ICF will not be replenished. Albumin is slow to be excreted and the transfused volume is excreted much slower than with replacement solutions. Plasma protein solutions are expensive and supply is limited. The fluid is initially excreted because of a fall in ADH level falling stimulation of the volume receptors.

8.2: Infusion of Hypertonic Saline

Hypertonic saline 3% has an osmolality (about 900 mosm/l) three times that of plasma. The fluid shifts & osmolar changes that occur with its infusion can be predicted.

Water crosses cell membranes easily and distributes passively in response to osmolar gradients. The Na+ content of the fluid limits the distribution of the infused fluid to the ECF. The hypertonic solution will also draw water out of cells decreasing intracellular fluid volume.

As an example, consider a rapid infusion of 1,000 mls of 3% saline into a 70kg subject with a total body water of 42 liters (ICF: 23 litres, ECF: 19 litres).

Just Before the infusion:

$\text{Total body solute content} = 42 \times 290 = 12,180 mOsm$
$\text{ECF solute content} = 19 \times 290 = 5,510 mOsm$
$\text{ICF solute content} = 23 \times 290 = 6,670 mOsm$

Immediately After the infusion :

$\text{Total body water} = 42 + 1 = 43 liters$

$\text{Total body solute content} = 12,180 + 900 = 13,080 mOsm$

$\text{ECF solute content} = 5,510 + 900 = 6,410 mOsm$

$\text{ICF solute content} = 6,670 mOsm$ (ie unchanged)

The prediction is:

$\text{Final osmolarity} = \frac{13{,}080}{43} = 304 \frac{mOsm}{l}$

$\text{ECF volume} = \frac{6{,}410}{304} = 21.1 litres$

$\text{ICF volume} = \frac{6{,}670}{304} = 21.9 litres$

Is the increase in osmolality enough to be sensed by the osmoreceptors?

Yes. The increase in ECF volume is 2.1 litres with about a quarter of this (say 500 mls) intravascularly. Plasma osmolality has increased by 4.8% and this is well above the threshold (1 to 2%) of the hypothalamic osmoreceptors.

Is the increase in blood volume enough to be sensed by the low pressure (volume) baroreceptors?

Yes. The blood volume has increased by about 10%. The volume receptors respond to changes above about 7 to 10%.

The increase in osmolality will be sensed by the osmoreceptors in the hypothalamus and this will be a potent stimulus to the secretion of ADH to retain water in the kidneys. Thirst will also be increased. The increase in blood volume is at about the lower level of sensitivity of the volume receptors. The effect via the volume receptors will be to inhibit ADH secretion to allow water excretion. In general, volume stimuli tend to be less sensitive but more potent than osmotic stimuli.

There will also be effects on Na^+ excretion. The volume expansion will stimulate secretion of atrial natriuretic factor (ANF). Secretion of aldosterone will be inhibited because of a decreased renin and angiotensin II production. ANF also inhibits renin secretion.

The final outcome of all these changes is natriuresis and excretion of the excess water. The increased osmolality causes an increased ADH and this will tend to inhibit the rate of excretion of the excess water.

The decrease in ICF volume may have effects on the brain causing confusion and mental obtundation due to cerebral cellular dehydration and hypertonicity. These effects on cerebral function will probably be the predominant clinical effects. The function of other organs or tissues in unlikely to be significantly affected.

The increase in ISF volume is not sufficient to cause oedema or interfere with gas transfer or nutrient and waste transfers between cells and capillaries.

8.3: Infusion of Hydrochloric Acid

At first this seems like a silly thing to do, but intravenous infusions of hydrochloric acid are sometimes used in Intensive Care Units in patients with chronic respiratory acidosis and high plasma bicarbonate levels as a way to more rapidly return the bicarbonate towards normal levels.

As an example, consider the infusion via a central line of 100 mls of 1N hydrochloric acid solution in a healthy adult. This represents an acute acid load of 100 mmols of H^+ which is sufficient to cause a metabolic acidosis. The defence against changes in $[H^+]$ involves buffering, compensation and correction.

Buffering

Buffering is a rapid physicochemical process that involves titration of the acid by the body's extracellular buffers (predominantly bicarbonate). Assuming a $[HCO_3^-]$ of 24 mmols/l and an extracellular volume of 19 liters, this represents a bicarbonate pool in ECF of about 450 mmols. An acid load of 100 mmols of $[H^+]$ will titrate the bicarbonate buffer to about 18.7 mmol/l (ie $\frac{350}{450} \times 24$) assuming all the buffering is by bicarbonate.

Compensation

The metabolic acidosis will stimulate the peripheral chemoreceptors resulting in an increase in ventilation. The resultant hypocapnia is the physiological compensatory response which returns pH towards normal. This response starts early but can take 12 to 24 hours to reach its maximum value. Compensation will not return pH completely to normal. The expected pCO_2 at maximum compensation is:

Expected $pCO_2 = 1.5 \times [HCO_3^-] + 8$

where pCO_2 is arterial pCO_2 in mmHg and $[HCO_3^-]$ is arterial bicarbonate (in mmol/l) calculated from arterial blood gases.

Correction

The kidney will excrete the excess acid anion (Cl^-) and this is equivalent to reabsorption of bicarbonate & excretion of acid. Normal acid-base status will be restored.

Other Physiological Effects.

These include:

- The oxygen dissociation curve will be shifted to the right by the acidosis. This decrease in oxygen affinity will assist peripheral oxygen unloading. Subsequently, the acidosis causes a decrease in 2,3 DPG synthesis and the ODC moves leftward
- Anion gap will be unchanged and the acidosis will tend to be a hyperchloraemic metabolic acidosis
- Metabolic acids do NOT cross the blood-brain barrier so direct effects on the brain are not significant. (As mentioned above, the respiratory centre will be stimulated secondary to stimulation of the peripheral chemoreceptors)
- Hyperkalaemia occurs due to H^+-K^+ exchange across cell membranes and urinary K^+ losses are increased. (Hyperkalaemia is less common when the metabolic acid involves organic anions -eg lactate- as the anion tends to cross cell membranes with the H^+ and the net cellular exchange with K^+ is less).

8.4: Infusion of 8.4 Percent Sodium Bicarbonate Solution

As an example, consider the infusion of an 8.4% $NaHCO_3$ solution. This is a molar solution of $NaHCO_3$. Dissociation into two particles in solution results in a solution with an osmolality of 2,000 mOsm/kg. This is about 7 times the plasma osmolality!

Infusion of this solution has effects because it is:

- Hypertonic (2,000 mOsm/l) with a high $[Na^+]$
- Alkalinising (HCO_3^- load).

The high sodium concentration restricts the distribution of the solution to the ECF. The hypertonic nature of the solution draws water out of cells until the ECF and ICF tonicities are equal. The increase in ECF volume will be greater than the volume of solution administered into it.

The ECF $[Na^+]$ will increase dependent on the amount of solution administered but the water drawn out of the cells will tend to minimise this increase. Sodium bicarbonate solution has occasionally been recommended for emergency treatment of acute hyponatraemia particularly where there was also a perceived benefit of the alkalosis.

The ECF $[HCO_3^-]$ will increase and this exogenous administration of base will cause a metabolic alkalosis. This causes intracellular movement of K^+ and ECF $[K^+]$ will decrease. This is the basis of the use of $NaHCO_3$ solution for the emergency treatment of hyperkalaemia.

Under normal circumstances, if the plasma bicarbonate rises above about 27 mmol/l then HCO_3^- is rapidly excreted in the urine. A metabolic alkalosis will rapidly correct unless there is some additional factor which maintains it. Because of the brief nature of the alkalosis, the compensatory hypoventilation is minimal.

There are conflicting influences on ADH levels:

- A rise in extracellular tonicity of 1 to 2% or more will increase ADH levels (effect via hypothalamic osmoreceptors)
- An increase in blood volume of 7 to 10% or more will decrease ADH levels (effect via low pressure baroreceptors).

A decrease in ADH will increase water excretion. An increase in blood volume due to $NaHCO_3$ infusion will cause a fall in plasma oncotic pressure and water reabsorption in the proximal tubule will decrease slightly due to glomerulotubular imbalance.

The increases in tonicity and blood volume can be estimated from a knowledge of the volume of solution administered.

[See similar calculations in Section 8.2.]

8.5: Infusion of Hypertonic Mannitol Solutions

8.5.1: Why use Mannitol?

Hypertonic mannitol solutions are used clinically for:

- Cerebral dehydration - to decrease an elevated intracranial pressure
- Renal Protection - to protect against development of renal failure due to osmotic diuresis in some clinical situations (eg with rhabdomyolysis)

The hypertonicity causes passive movement of water across lipid barriers in response to the osmotic gradient.

8.5.2: Effects of Mannitol Infusion

Mannitol is a monosaccharide which is easy to produce and stable in solution. It is used clinically in doses ranging from 0.25 to 1.5 g/kg body weight. Solutions of 10% mannitol (osmolality 596 mOsm/kg) and 20% mannitol (osmolality 1,192 mOsm/kg) are commonly available for clinical use.

Cerebral effects

Mannitol does not cross the blood brain barrier so an elevated plasma osmolality due to a infusion of hypertonic mannitol is effective in removing fluid from the brain. This is called 'mannitol osmotherapy'. In the past, other hypertonic solutions (eg hypertonic urea solution) have been used and currently in some places hypertonic glycerol solutions are available as an alternative to mannitol.

Mannitol infusions are useful to acutely decrease elevated intracranial pressure due to an intracranial space occupying lesion. A typical use would be in a patient with an intracerebral haematoma due to an acute traumatic head injury. The effect is rapid in onset (minutes) but only temporary (as the mannitol is excreted) but its use buys time for urgent definitive therapy (eg surgical evacuation of the haematoma and surgical haemostasis). A typical dose in an adult would be 0.5-1.5g/kg administered as the 20% solution.

Repeated doses of mannitol have less effect and as some slowly enters the brain, rebound intracranial hypertension is a risk. As the blood-brain barrier is probably disrupted in damaged areas of the brain, mannitol may be both less effective here and also more may enter the brain at these places. However, the therapeutic effect of mannitol is not dependent on a specific action at damaged areas of the brain but rather on a global effect in decreasing intracranial fluid volume and intracranial pressure so this has little relevance for a first dose of mannitol and especially if definitive surgical treatment is successful. Much more problematical is use of repeated doses of mannitol in ICU patients with traumatic intracranial hypertension in whom there is no surgically correctable cause; such use is usually futile.

The brain cells also compensate for the presence of continued hypertonicity by the intracellular production of '**idiogenic osmoles**'. The effect is to increase intracellular tonicity and allow brain cell volume to return towards normal presumably with improvement of intracellular functions despite the continued hypertonicity.

Use of mannitol infusions is common intraoperatively in some neurosurgical procedures. The aim is to decrease intracranial pressure and produce a 'slack brain' to facilitate surgical access.

Mannitol does not cross cell membranes so the cell volume of most other cells in the body is also decreased.

Renal effects

In the renal glomeruli, mannitol is freely filtered. It is not secreted or reabsorbed by the tubules. In the doses used clinically it retains water with it in the tubule and causes an 'osmotic diuresis'. Consequently, mannitol is classified as an 'osmotic diuretic'. The high flow of retained tubule fluid tends to have a flushing effect and washes fluid and solutes from the kidney. This effect is useful clinically in management of rhabdomyolysis. The aim is to 'wash' the myoglobin out of the tubules and prevent it precipitating there with obstruction and development of acute renal failure. The effect of mannitol for this use is aided by maintenance of adequate intravascular volume and by urinary alkalinisation (by administration of IV sodium bicarbonate).

Intravascular volume effects

Attention to intravascular volume status is important during any clinical use of mannitol. Initially, the tissue dehydrating effect will increase intravascular volume with the risk of precipitating volume overload and hypertension and/or acute congestive heart failure. Subsequently, the diuretic effect may result in hypovolaemia (and hypernatraemia). Frusemide (a loop diuretic) may be a useful adjunct in some cases to minimise the initial hypervolaemia.

Other effects

The increased intravascular water volume decreases the red cell concentration (decreased haematocrit) with a resultant decrease in blood viscosity. This may improve flow and oxygen delivery to some areas.

Mannitol has free radical scavenging properties and these may contribute to its therapeutic effects (though this has not so far been established).

Effects of Mannitol

Osmotic Effects (due hypertonicity)

- Intracellular dehydration
- Expansion of ECF volume (except brain ECF)
- Haemodilution
- Diuresis due osmotic effects and ECF expansion

Non-Osmotic Effects

- Decreased blood viscosity (with improved tissue blood flow)
- Possible Cytoprotective effect (due free radical scavenging)
- Cardiovascular effects secondary to expanded intravascular volume (eg increased cardiac output, hypertension, heart failure, pulmonary oedema)

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